KAYA Frame Grabbers
Feature Guide

October 2026
Rev 2026.1.3



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Revision History

Version	Date	Notes
2026.1.3	10.2026	Initial release

Table 1 – Revision History

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2 Introduction

2.1 Safety Precautions

Please take the time to read through the precautions listed below in order to prevent preventable and unnecessary injuries and damage to you, other personnel or property. Read these safety instructions carefully prior to your first use of the product, as these precautions contain safety instructions that must be observed. After reading through this manual, be sure to follow it to prevent misuse of product.

⚠ Caution! Read Carefully and do not disregard these instructions.
In the event of a failure, disconnect the power supply Disconnect the power supply immediately and contact our sales personnel for repair. Continuing to use the product in this state may result in a fire or electric shock.
If an unpleasant smell or smoking occurs, disconnect the power supply. Disconnect the power supply immediately! Continuing to use the product in this state may result in a fire or electric shock. After verifying that no smoking is observed, contact our sales personnel for repair.
Do not disassemble, repair or modify the product. This may result in a fire or electric shock due to a circuit shortage or heat generation. Contact our sales personnel prior to inspection, modification or repair.
Do not place the product on unstable surfaces. Otherwise, it may drop or fall, resulting in injury to persons or the camera.
Do not use the product if dropped or damaged. Otherwise, a fire or electric shock may occur.
Do not touch the product with metallic objects. Otherwise, a fire or electric shock may occur.
Do not place the product in dusty or humid environments, nor where water may splash. Otherwise, a fire or electric shock may occur.
Do not wet the product or touch it with wet hands. Otherwise, the product may fail or it may cause a fire, smoking or electric shock.
Do not touch the gold-plated sections of the connectors on the product. Otherwise, the surface of the connector may be contaminated by sweat or skin-oil, resulting in contact failure of a connector, malfunction, fire or electric shock due to static electricity discharge.
Do not use or place the product in the following locations. ▪ Unventilated areas such as closets or bookshelves. ▪ Near oils, smoke or steam. ▪ Next to heat sources. ▪ A closed (and not running) car where the temperature becomes high. ▪ Static electricity replete locations ▪ Near water or chemicals. Otherwise, a fire, electric shock, accident or deformation may occur due to a short circuit or heat generation.
Do not place heavy objects on the product. Otherwise, the product may be damaged.
Be sure to discharge static electricity from body before touching any sensitive electronic components. The electronic circuits in your computer and the circuits on the camera and the ***Frame grabber*** board are sensitive to static electricity and surges. Improper handling may seriously damage the circuits. In addition, do not let your clothing come in contact with the circuit boards or components. Otherwise, the product may be damaged.

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2.2 Disclaimer

KAYA Instruments will assume no responsibility for any damage that may ensue by the use of this product for any purpose other than intended, as previously stated. Without detracting from what was previously written, please be advised that the company will take no responsibility for any damages caused by:

- Earthquake, thunderstrike, natural disasters, fire caused by use beyond our control, willful and/or accidental misuse and/or use under other abnormal and/or unreasonable conditions.
- Secondary damages caused by the use of this product or its unusable state (business interruption or others).
- Use of this product in any manner that contradicts this manual or malfunctions that may occur due to connection to other devices. Damage to this product that is out of our control or failure due to modification.
- Accidents and/or third parties that may be involved.

Additionally, KAYA Instruments assumes no responsibility or liability for:

- Erasure or corruption of data caused by the use of this product.
- Any consequences or other abnormalities following the use of this product.

Repairs to this product are carried out by replacing it on a chargeable basis and not by repairing the faulty device. Non-chargeable replacement is offered for initial failure, as long as it is reported no later than two weeks post-delivery of the product.

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3 Overview

3.1 Document Scope

The purpose of this document is to describe the provided functionality and features of KAYA's Frame Grabbers. Camera connectivity and streaming can be easily achieved in few easy steps with almost no configurations. Camera control is provided through standard Gen<I>Cam interface subordinate to camera's descriptive schema (xml) file. Advanced features and custom configurations can be done to enhance streaming and image processing of camera output. These are available using interactive Gen<I>Cam interface and provided API functionality. SDK functionality is subject to hardware device and burned firmware capabilities. A firmware and software upgrade may be needed to support complete functionality set.

For more information about API functionality and SDK usage please refer to Vision Point II API Data Book. The parameters described in this document are Frame Grabber, Local Device and Stream parameters and can be accessed from API using provided KYVPParametersHandler library.

3.2 Document Structure

This Frame Grabber Feature Guide is divided into few major topics each related to different functionalities:

- PCI Interface control
- Local Device control
- Stream control

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4 PCI Interface control

The PCI Interface represents a single physical interface – a Frame Grabber. The Frame Grabber configurations contain various features and controls. Hardware information, stream state, I/O definition, and more can be modified using the standard Gen<i>Cam interface.
Different boards may include different feature sets.

4.1 Hardware Information

The Hardware information contains parameters describing the currently connected hardware device and includes device capabilities, basic connectivity details and currently running firmware. This information can be used to identify the specific card and its capabilities and inform if a firmware update is needed to support complete functionality set. It can also help to indicate about certain performance issues. The hardware information parameters can be found in the following table:

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam	
Gen<i>Cam Category: Hardware Information						
Firmware Version	Version of the firmware in the PCI Interface	DeviceFirmwareVersion	String			
Firmware status	Status of the firmware in the PCI Interface	DeviceFirmwareStatus	String			
Software status	Compatibility of the running software version with the running firmware	DeviceSoftwareStatus	String			
Device Serial Number	Serial Number of the PCI Interface	DeviceSerialNumber	String			
Revision	Revision of the PCI Interface	DeviceRevision	Integer			
Maximum available Links	Maximum available links on the PCI Interface	DeviceMaxLinks	Integer			
Physical links	Maximum physical links available on the PCI Interface	DevicePhysicalLinksMax	Integer			
Stream channels	Maximum stream channels available on the PCI Interface	DeviceMaxStreamChannels	Integer			
Min. PCIe generation required	Minimal device requirement for PCIe generation	DevicePciGenerationMin	Integer			
Min. PCIe lanes required	Minimal PCI Interface requirement for number of PCI lanes	DevicePciLanesMin	Integer			
PCIe Generation	PCIe generation configuration	DevicePciGeneration	Integer			
PCIe lanes	Maximum available lanes on the PCI Interface PCIe slot	DevicePciLanes	Integer	1,2,4,8		
Core Temperature	PCI Interface core temperature in Celsius	DeviceTemperature				Temperature is in °C
Status	Status of the PCI Interface	DeviceStatus	String			

Table 2 – Hardware information parameters

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4.2 Device Control

4.2.1 Discovery

KAYA's Frame Grabbers API provides different camera discovery modes. By default, the Normal discovery mode is active which includes camera negotiation, reset sequence and setting the camera's default speed and topology. The different camera discovery modes are available by setting existing configurations. Manual control and automatic management of PoCXP are provided for CoaXPress cameras which draw power via coax cables, instead of external power supply.

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: Device Control/Discovery						
Camera Discovery Verify Speed	Camera discovery verify speed	CameraDiscoveryVerifySpeed	Boolean			
Camera Command Timeout	Indicates the command timeout of all links in microseconds(us). This corresponds to the maximum response time of the device for a command sent	DeviceLinkCommandTimeout	Float			
Camera Discovery Links	Non zero value will be used as bitmask to define grabber links participating in for camera detection. If left 0 then all links will be used	CameraDiscoveryLinks	Integer	0 255	Min Max	
Link 0	Perform detection on link 0	CameraDiscoveryLink0	Boolean			See Note [1]

Table 3 – Device Control parameters

NOTE:

1. The table above shows the parameters for one link 0 as an example. If multiple links are configured, an identical set of parameters is displayed for each link, supporting up to eight links.

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4.2.2 General

KAYA Frame Grabbers include a Timestamp mechanism for tagging frames and I/O events. Timestamp parameter reflects a global counter value, in nanoseconds. The counter value is represented by 64bit unsigned integer which wraps around when maximum value is reached and can be read from the “Timestamp” register. “TimestampReset” command force resets the timestamp counter to 0.

The timestamp counter may not be stopped but the “TimestampLatch” may capture the counter value in the moment it is issued. The captured value will be stored in 64bit unsigned integer register “TimestampLatchValue” until the next “TimestampLatch” command is issued. The timestamp parameters are summarized in the following table:

KAYA Frame Grabbers parameters are summarized in the following table:

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: DeviceControl						
Device User ID	User-programmable device identifier	DeviceUserID	String			
Save 'Device User ID'	Writes 'Device User ID' to hardware	DeviceUserIDWrite	Command	1 - Activate		
Device reset	Reset the device	DeviceReset	Command	1 - Activate		
Device stream control reset	Reset the device's stream control	DeviceStreamControlReset	Command	1 - Activate		
Timestamp Counter	Reports the current value of the device timestamp counter.	Timestamp	Integer (8 bytes)			Value in nanoseconds
Timestamp Reset	Resets the current value of the device timestamp counter	TimestampReset	Command	1 - Activate		
Timestamp Latch	Latches the current timestamp counter into Timestamp Latch Value	TimestampLatch	Command	1 - Activate		
Timestamp Latch value	Reports the Latched value of the timestamp counter.	TimestampLatchValue	Integer (8 bytes)			Value in nanoseconds

Table 4 – Timestamp parameters

4.3 PoCXP Automatic Management

Power over CoaXPress (PoCXP) is a feature that allows the transmission of power along with data over coaxial cables. This technology is part of the CoaXPress standard. The PTU (Power Transmitting Unit) is responsible for delivering power to the device safely and in compliance with the CoaXPress standard.

KAYA's software stack continuously monitors the CoaXPress connection state and automatically enables or disables PoCXP as needed.

When supported, the frame grabber can power a connected camera in the background, even if KAYA Vision Point, KAYA Vision Studio, or any other KAYA API-based application is not running.
This enables faster and more seamless connection to compatible CoaXPress cameras.

Automatic PoCXP management is the recommended method for controlling PoCXP, as it eliminates the need for manual power management and ensures that camera power is applied only when required.

If you require more precise control over when PoCXP is enabled or disabled, refer to the following sections, which describe the available PoCXP control commands.

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: DeviceControl/PoCXP Control						
PoCXPAuto available	Indicates whether PoCXP can be controlled automatically	PoCXPAutoAvailable	Enumeration	0 1	PoCXPAutoAvailableOff PoCXPAutOn	
PoCXP Auto monitoring active	Activates / deactivates automatic PoCXP monitor	PoCXPActive	Boolean			
PoCXP 0 control	Frame Grabber PoCXP channel 0 control	PoCXP0	Enumeration	0 1	PoCXPOff PoCXPOn	See Notes [1, 2]
PoCXP Host Connection Selector	PoCXP connection selector	CxpPoCxpHostConnectionSelector	Enumeration (Selector)	-1 0 1 2 3 4 5 6 7	All Link0 Link1 Link2 Link3 Link4 Link5 Link6 Link7	
PoCXP Auto	Activate automaticcontrol of the Power over CoaXPress (PoCXP) for the Link	PoCXPAuto	Command			
PoCxp Turn Off	Disable Power over CoaXPress (PoCXP) for the Link	CxpPoCxpTurnOff	Command			
PoCXP Status	Returns the Power over CoaXPress (PoCXP) status of the Device link	CxpPoCxpStatus	Enumeration	-1 0 1 2	Mixed Auto Off Tripped	Mixed statuses Automatically managed Forced Off Tripped
PoCXP Voltage 0		PoCXP_ADCVoltageReadout0	Float			See Note [2]
PoCXP Current 0		PoCXP_ADCCurrentReadout0	Float			See Note [2]

Table 5 – Automatic PoCXP control parameters

NOTE:

1. The legacy Grabber parameters "PoCXP0" ... "PoCXP7" are still available when automatic PoCXP is active, but they become read-only in this case. The values of those parameters can be read to get the current state of PoCXP on each channel.
2. The table above shows the parameters PoCXP Voltage 0 and PoCXP Current 0 for one link 0 as an example. If multiple links are configured, an identical set of parameters is displayed for each link, supporting up to eight links.

4.3.1 Feature Availability

Automatic PoCXP power management requires compatible hardware, firmware, and software.

To determine whether the feature is available in your system, read the PoCXPAutoAvailable frame grabber parameter.

- Yes – Automatic PoCXP power management is supported.
- No – The current hardware, firmware, software, or camera configuration does not support this feature.

4.3.2 Channel Selection

The CxpPoCxpTurnOff, CxpPoCxpAuto, and CxpPoCxpStatus parameters follow the Gen<i>Cam SFNC standard with the following KAYA-specific extension:

The CoaXPress channels affected by these commands are determined by the current value of CxpPoCxpHostConnectionSelector.

- If CxpPoCxpHostConnectionSelector = -1, the command applies to all available CoaXPress channels.
- If CxpPoCxpHostConnectionSelector is set to a specific channel number, the command applies only to the selected channel.

This allows PoCXP power management to be controlled globally or independently for each CoaXPress channel.

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4.3.3 Fan Control

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: Device Control / FanControl						
PWM frequency	Fan control clock frequency in KHz	PWM_frequency	Float			In units of KHz
PWM duty cycle	Fan duty cycle in percentage to PWM period	PWM_duty_cycle	Integer	0 –100		In units of %
FAN On Threshold	When the card's temperature rise above the max threshold it will turn the fan On	PWM_max_temp	Float			
FAN Off Threshold	When the card's temperature drops below the min threshold it will turn the fan Off	PWM_min_temp	Float			

Table 6 – Fan control parameters

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4.4 Bandwidth Test

KAYA's Frame Grabbers also include a Bandwidth test feature for PCIe throughput testing. This feature allows the user to set the desired buffer size and execute a bandwidth test for currently selected Frame Grabber. This test should be performed as a standalone test, without any streaming during its execution. Succession of the bandwidth test will result in "Finished" status while abortion of the test will result in "Stopped" status. During the test the status will change to "Running". The bandwidth test parameters are summarized in the next table:

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: DeviceControl/ BandwidthTest						
Reference buffer size (Bytes)	Size in Bytes of reference buffer used for bandwidth test	BandwidthTestReferenceBufferSize	Integer			
Payload buffer width	Width of payload buffer(s) used for bandwidth test	BandwidthTestPayloadBufferWidth	Integer			
Payload buffer height	Height of payload buffer(s) used for bandwidth test	BandwidthTestPayloadBufferHeight	Integer			
Payload buffer size (MB)	Size in MB of payload buffer(s) used for bandwidth test	BandwidthTestPayloadBufferSize	Integer			
Payload buffers count	Number of buffers used for bandwidth test	BandwidthTestPayloadBuffersCount	Integer			
Perform bandwidth test	Perform bandwidth test	BandwidthTestPerform	Command	1 - Activate		
Abort bandwidth test	Abort bandwidth test	BandwidthTestAbort	Command	1 - Activate		
Bandwidth test status	Current status of bandwidth test	BandwidthTestStatus	Enumeration	0	Stopped	
				1	Started	
				2	Running	
				3	Finished	
				4	Error	
Average transfer time (usec)	Average time for payload buffer(s) transfer used for bandwidth test	BandwidthTestAverageTime	Float			
Average transfer speed (MB/sec)	Average speed of payload buffer(s) transfer used for bandwidth test	BandwidthTestAverageSpeed	Float			

Table 7 – Bandwidth test parameters

4.5 I/O Control

KAYA's Frame Grabber boards include a large array of auxiliary signal which can be used to initiate on-board events, transmitted to other devices or rerouted from other signals, such as CoaXPress triggers and GPIO's. Additionally, these auxiliary signals can be used to communicate with complex devices, such as encoders, strobe controls and drive controls.

The GPIOs can be configured and controlled from the provided API and be set as a trigger sources. The API enables routing of any input to any output as well as to the CoaXPress IO and Trigger lines. The I/O controller is responsible for the following features:

- Triggers
- I/O lines
- Encoders
- Timers

4.5.1 Digital I/O Control

4.5.1.1 Line Selector

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: I/O Control/Digital I/O Control						
Line Selector	Selects the physical pin of the external camera connectors	LineSelector	Enumeration			See Table 9 – Line Selector selection options
Line Mode	Controls if the physical Line is used to Input or Output a signal	LineMode [LineSelector]	Enumeration	0 1	Input Output	Table 9
Line Inverter	Controls the inversion of the signal of the selected input or output Line.	LineInverter [LineSelector]	Boolean	0 1	False True	
Line Source	Selects which internal acquisition or I/O source signal to output on the selected Line	LineSource [LineSelector]	Enumeration			See section Link Trigger Source options
Line Status	Returns the current status of the selected input or output Line	LineStatus [LineSelector]	Boolean	0 1	False True	
Line Format	Controls the current electrical format of the selected physical input or output Line	LineFormat [LineSelector]	Enumeration	0 1 2 3 4 5 6 7	NoConnect TriState TTL LVDS RS422 OptoCoupled LVTTL NonAvailable	
Line Event Mode	Selects the line event generation mode	LineEventMode [LineSelector]	Enumeration	0 1 2 3	Disabled RisingEdge FallingEdge AnyEdge	
Line Status All	Returns the current status of all available Line signals at time of polling in a single bitfield	LineStatusAll	Integer			Value for all lines, not effected by line selector value

Table 8 – Line Selector parameters

Value	Output	Gen<i>Cam parameter name
00	OptoCoupled Input 0	KY_OPTO_IN_0
01	OptoCoupled Input 1	KY_OPTO_IN_1
02	OptoCoupled Input 2	KY_OPTO_IN_2
03	OptoCoupled Input 3	KY_OPTO_IN_3
04	OptoCoupled Input 4	KY_OPTO_IN_4
05	OptoCoupled Input 5	KY_OPTO_IN_5
06	OptoCoupled Input 6	KY_OPTO_IN_6
07	OptoCoupled Input 7	KY_OPTO_IN_7
08	LVDS Input 0	KY_LVDS_IN_0
09	LVDS Input 1	KY_LVDS_IN_1
0A	LVDS Input 2	KY_LVDS_IN_2
0B	LVDS Input 3	KY_LVDS_IN_3
0C	TTL 0	KY_TTL_0
0D	TTL 1	KY_TTL_1
0E	TTL 2	KY_TTL_2
0F	TTL 3	KY_TTL_3
10	TTL 4	KY_TTL_4
11	TTL 5	KY_TTL_5
12	TTL 6	KY_TTL_6
13	TTL 7	KY_TTL_7
14	LVTTL 0	KY_LVTTL_0
15	LVTTL 1	KY_LVTTL_1
16	LVTTL 2	KY_LVTTL_2
17	LVTTL 3	KY_LVTTL_3
18	LVTTL 4	KY_LVTTL_4
19	LVTTL 5	KY_LVTTL_5
1A	LVTTL 6	KY_LVTTL_6
1B	LVTTL 7	KY_LVTTL_7
1C	OptoCoupled Output 0	KY_OPTO_OUT_0
1D	OptoCoupled Output 1	KY_OPTO_OUT_1
1E	OptoCoupled Output 2	KY_OPTO_OUT_2
1F	OptoCoupled Output 3	KY_OPTO_OUT_3
20	OptoCoupled Output 4	KY_OPTO_OUT_4
21	OptoCoupled Output 5	KY_OPTO_OUT_5
22	OptoCoupled Output 6	KY_OPTO_OUT_6
23	OptoCoupled Output 7	KY_OPTO_OUT_7
24	LVDS Output 0	KY_LVDS_OUT_0
25	LVDS Output 1	KY_LVDS_OUT_1
26	LVDS Output 2	KY_LVDS_OUT_2
27	LVDS Output 3	KY_LVDS_OUT_3

Table 9 – Line Selector selection options

Large array of GPIO is available for configuring trigger source from external signal generators, such as TTL, LVDS, LVTTL and OptoCoupled. The auxiliary GPIO signals can be used to initiate on-board events, transmitted to other devices or rerouted from other signals, such as CoaXPress triggers and other GPIOs.

The GPIOs can be configured from the API and be set as a trigger sources. The API enables routing of any input to any output as well as to the CoaXPress IO and Trigger lines. For complete GPIO pinout, see relevant Frame Grabber datasheet.

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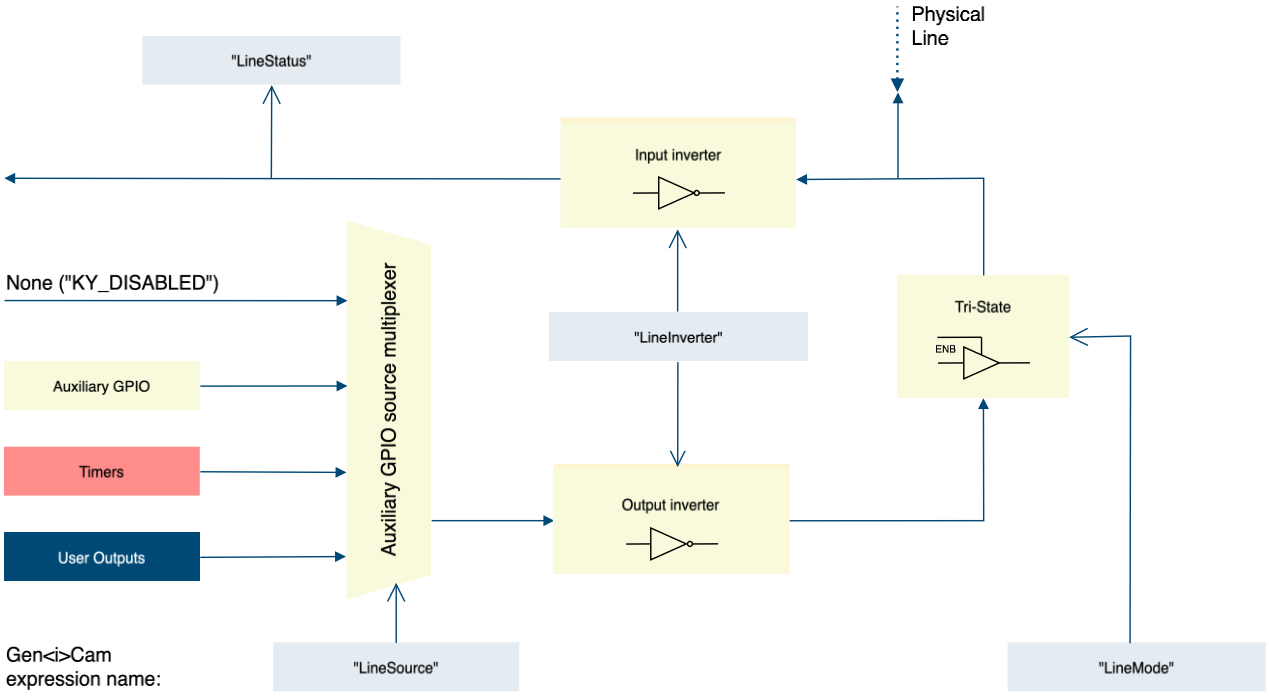


Figure 1 – Digital I/O Line structure

4.5.1.2 User Output Selector

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: I/O Control/Digital I/O Control						
User Output Selector	Selects which bit of the user output register will be set by UserOutputValue	UserOutputSelector	Enumeration	UserOutput0 UserOutput1 UserOutput2 UserOutput3 UserOutput4 UserOutput5 UserOutput6 UserOutput7		
User Output Value	Sets the value of each of the User Output register	UserOutputValue	Boolean			
User Output Value All	Sets the value of all the bits of the User Output register. It is subject to the UserOutputValueAllMask	UserOutputValueAll	Integer			
User Output Value All Mask	Sets the write mask to apply to the value specified by UserOutputValueAll before writing it in the User Output register. If the UserOutputValueAllMask feature is present, setting the user Output register using UserOutputValueAll will only change the bits that have a corresponding bit in the mask set to one	UserOutputValueAllMask	Integer			

Table 10 – User Output Selector parameters

4.5.2 Encoder Control

4.5.2.1 Encoder Selector

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: I/O Control/Encoder Control						
Encoder Output Mode	Selects trigger output signal behaviour	EncoderOutputMode [EncoderSelector]	Enumeration	0 1 2 3 4	Disabled Position Anystep Stepforward Stepbackward	
Encoder Inverter	Controls the inversion of the signal of the selected encoder	EncoderInverter [EncoderSelector]	Boolean	0 1	False True	
Encoder A Source	Encoder source selection for A input	EncoderASource [EncoderSelector]	Enumeration			See Table 12 – Encoder A/B Source selection options
Encoder B Source	Source I/O B	EncoderBSource [EncoderSelector]	Enumeration			See Table 12 – Encoder A/B Source selection options
Encoder Value	Encoder value in step counts	EncoderValue [EncoderSelector]	Integer			Writing this register will pre-set the count
Encoder Position	The value to compare with the Encoder Position Value	EncoderPosition [EncoderSelector]	Integer			Only used if "EncoderOutputMode " is set to "Position"
Encoder Filter	Filter for encoder signals	EncoderFilter [EncoderSelector]	Float			In units of microseconds (us) 8ns resolution using fraction value
Encoder Reset Activation	Activation mode of encoder reset signal	EncoderResetActivation [EncoderSelector]	Enumeration	0 1 2 3 4	RisingEdge FallingEdge AnyEdge LevelHigh LevelLow	
Encoder Reset Source	Source I/O for encoder reset	EncoderResetSource [EncoderSelector]	Enumeration			See section Link Trigger Source options
Encoder Reset	Software reset for encoder	EncoderReset [EncoderSelector]	Command	1 - Activate		
Encoder Value at Reset	Last position counter before encoder rest	EncoderValueAtReset [EncoderSelector]	Integer			
Encoder Event Mode	Enables event generation for encoder	EncoderEventMode [EncoderSelector]	Enumeration	0 1	Disable Enable	

Table 11 – Encoder Control parameters

Value	Output	Gen<i>Cam parameter name
00	Disabled	KY_DISABLED
01	OptoCoupled Input 0	KY_OPTO_IN_0
02	OptoCoupled Input 1	KY_OPTO_IN_1
03	OptoCoupled Input 2	KY_OPTO_IN_2
04	OptoCoupled Input 3	KY_OPTO_IN_3
05	OptoCoupled Input 4	KY_OPTO_IN_4
06	OptoCoupled Input 5	KY_OPTO_IN_5
07	OptoCoupled Input 6	KY_OPTO_IN_6
08	OptoCoupled Input 7	KY_OPTO_IN_7

09	LVDS Input 0	KY_LVDS_IN_0
0A	LVDS Input 1	KY_LVDS_IN_1
0B	LVDS Input 2	KY_LVDS_IN_2
0C	LVDS Input 3	KY_LVDS_IN_3
0D	TTL 0	KY_TTL_0
0E	TTL 1	KY_TTL_1
0F	TTL 2	KY_TTL_2
10	TTL 3	KY_TTL_3
11	TTL 4	KY_TTL_4
12	TTL 5	KY_TTL_5
13	TTL 6	KY_TTL_6
14	TTL 7	KY_TTL_7
15	LVTTL 0	KY_LVTTL_0
16	LVTTL 1	KY_LVTTL_1
17	LVTTL 2	KY_LVTTL_2
18	LVTTL 3	KY_LVTTL_3
19	LVTTL 4	KY_LVTTL_4
1A	LVTTL 5	KY_LVTTL_5
1B	LVTTL 6	KY_LVTTL_6
1C	LVTTL 7	KY_LVTTL_7
40	TriggerN 0	TriggerN0
41	TriggerN 1	TriggerN1
42	TriggerN 2	TriggerN2
43	TriggerN 3	TriggerN3
44	TriggerN 4	TriggerN4
45	TriggerN 5	TriggerN5
45	TriggerN 6	TriggerN6
47	TriggerN 7	TriggerN7
48	TriggerN 8	TriggerN8
49	TriggerN 9	TriggerN9
4A	TriggerN 10	TriggerN10
4B	TriggerN 11	TriggerN11
4C	TriggerN 12	TriggerN12
4D	TriggerN 13	TriggerN13
4E	TriggerN 14	TriggerN14
4F	TriggerN 15	TriggerN15

Table 12 – Encoder A/B Source selection options

Configurable encoder triggers for both Shaft encoders and Quadrature Shaft encoders. Usually used to overcome image capture synchronization issues, by adjusting and controlling image acquisition using encoder physical steps rather than timed capture.

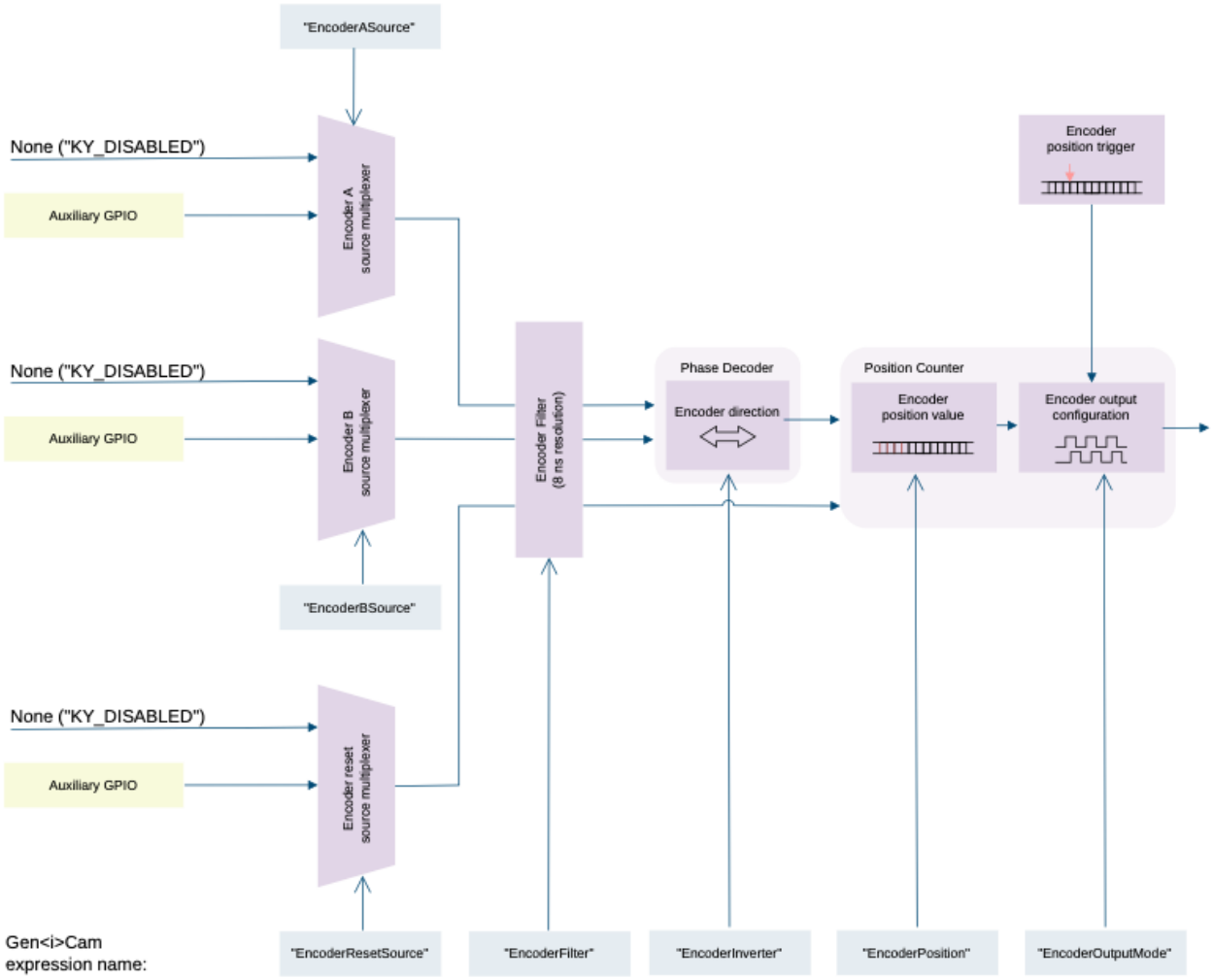


Figure 2 – Encoder triggers structure

While simple shaft encoders have one output, generating pulses according to step resolution, a quadrature shaft encoder has two outputs, called “A” and “B”, which are 90˚ out of phase. This allows interpreting the output of both lines to determine the direction of the encoder.

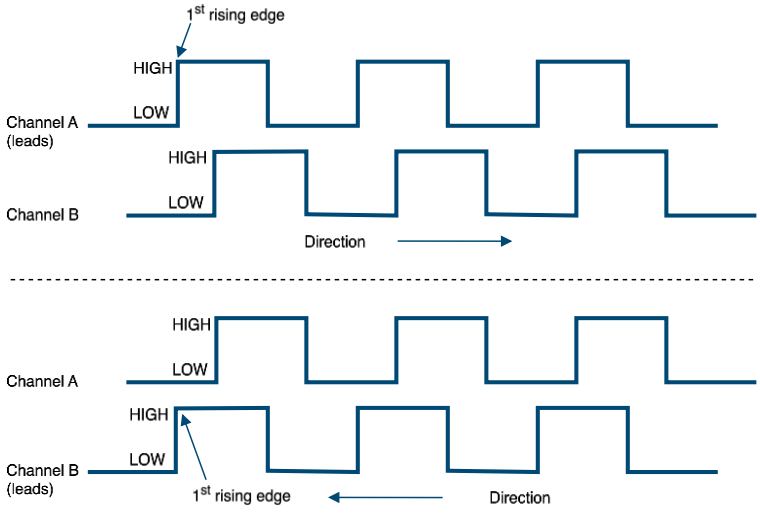


Figure 3 – Encoder channels

4.5.3 Encoder output mode

The encoder activation mode configures the capture criteria of trigger generation, according to encoder position and direction. Encoder direction output can be inverted, to define a downwards stepper. The different modes functionality is as follows:

1. Disabled: Signal pulse generation is disabled.
2. Position: Signal pulse generation will occur when value of Encoder Position will match the value of Position Trigger.
3. Any step: Signal pulse generation on every encoder step, regardless of encoder direction.
4. Step forward: Signal pulse generation every forward encoder step, backward step is ignored.
5. Step backward: Signal pulse generation every backward encoder step, forward step is ignored.

4.5.4 Encoder filter

The filter of the trigger signals acts as a de-bouncing mechanism for better handling generated noise. By default, the filter is disabled with the value of 0. The signal filter resolution can be set at 8ns intervals for high resolution functionality. If the trigger filter is set to a larger value than the width of the trigger pulse, then the pulse will be filtered out and no trigger will occur. Available interface in API provides input in microsecond; nevertheless, to achieve higher resolution, relevant fraction values should be entered after the decimal point.

4.5.5 Encoder Position Trigger and Encoder Position

Encoder Position Trigger defines the current encoder position while the Encoder Position defines the value which encoder step count should reach to issue the trigger. Writing to these registers will pre-set the count. The Encoder Position step counter depends on encoder resolution and is not bound by time limit.

4.5.6 Encoder Event Mode

Encode Event Mode may be enabled for selected encoder. This will generate event callback whenever such trigger is generated in hardware. Steps to enable and use such event mechanism are as follows:

Select encoder by setting the “EncoderSelector” parameter and enable “EncoderEventEnable” for selected encoder.

1. Register a callback function for Auxiliary events using KYVPLibExtension library KYVPExtension_PCIInterface_AuxDataCallback_Register() function.
2. To access the data attached to such event please refer to KYVP_AUX_DATA pointer in the callback.

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4.6 Link Trigger Control

4.6.1 Steps to properly configure Frame Grabber Triggers

1. "TriggerMode" is a grabber parameter subordinate to connected camera. Use the KYVPParametersHandler library function KYParametersHandler_SetValueEnumByValueName() to set the parameter value to "On".
2. The trigger source should be selected according to provided sources and available card GPIO. Only one source can be active, for each camera, at any time.
3. The Trigger Filter resolution ("TriggerFilter"), Activation Mode ("TriggerActivation") and Trigger Delay ("TriggerDelay") parameters should be configured according to desired output.
4. In some cases, the trigger sources should also be configured via provided API before trigger configuration is complete. (e.g if "KY_TIMER_ACTIVE_0" is to be selected as Camera Trigger source, then "Timer0" should first be configured as described in Timer Block configuration in this chapter).
5. After all configurations are complete, start the acquisition. At this point acquisition mechanism will wait for trigger, and Frame Grabber will acquire data upon trigger arrival.

4.6.1.1 Link Trigger Selector

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: I/O Control/Link Trigger Control						
Link Trigger Selector	Selects the trigger to configure	LinkTriggerSelector	Enumeration	Link 0 Link 1 Link 2 Link 3 Link 4 Link 5 Link 6 Link 7	Link0 Link1 Link2 Link3 Link4 Link5 Link6 Link7	
Link Trigger Mode	Controls if the selected trigger is active	LinkTriggerMode	Enumeration	0 1	Off On	
Link Trigger Activation	Specifies the activation mode of the trigger	LinkTriggerActivation	Enumeration	0 1 2 3 4 5	RisingEdge FallingEdge AnyEdge Rising Edge Invert Falling Edge Invert Any Edge Invert	
Link Trigger Source	Specifies the internal signal or physical input Line to use as the trigger	LinkTriggerSource	Enumeration			See section Link Trigger Source options
Link Trigger Delay	Specifies the delay in microseconds(us) to apply after the trigger reception before activating it	LinkTriggerDelay	Float	0.016000 – 34359738		In units of microseconds (us)
Link Trigger Filter	Filter for trigger, helps prevent signal de-bouncing. 8ns resolution, units in microseconds(us)	LinkTriggerFilter	Float	0 – 34359738		In units of microseconds (us)
Link Trigger Software	Generates an internal trigger. TriggerSource must be set to Software	LinkTriggerSoftware	Command	1 - Activate		
Link Trigger Event Enable	Enable/Disable event from link trigger on every activation	LinkTriggerEventEnable	Enumeration	0 1	Disabled Enabled	

Table 13 – Link Trigger Selector parameters

4.6.1.2 Link Trigger Source options

Enumerated below are available trigger sources for each trigger component. This is subject to device hardware, firmware and software capabilities.

Value	Source	Gen<i>Cam parameter name	I/O	Timer	Trigger	Encoder	Camera Trigger
0	Disabled	KY_DISABLED	✓	✓	✓	✓	✓
1	OptoCoupled Input 0	KY_OPTO_IN_0	✓	✓	✓	✓	✓
2	OptoCoupled Input 1	KY_OPTO_IN_1	✓	✓	✓	✓	✓
3	OptoCoupled Input 2	KY_OPTO_IN_2	✓	✓	✓	✓	✓
4	OptoCoupled Input 3	KY_OPTO_IN_3	✓	✓	✓	✓	✓
5	OptoCoupled Input 4	KY_OPTO_IN_4	✓	✓	✓	✓	✓
6	OptoCoupled Input 5	KY_OPTO_IN_5	✓	✓	✓	✓	✓
7	OptoCoupled Input 6	KY_OPTO_IN_6	✓	✓	✓	✓	✓

8	OptoCoupled Input 7	KY_OPTO_IN_7	✓	✓	✓	✓	✓
9	LVDS Input 0	KY_LVDS_IN_0	✓	✓	✓	✓	✓
10	LVDS Input 1	KY_LVDS_IN_1	✓	✓	✓	✓	✓
11	LVDS Input 2	KY_LVDS_IN_2	✓	✓	✓	✓	✓
12	LVDS Input 3	KY_LVDS_IN_3	✓	✓	✓	✓	✓
13	TTL 0	KY_TTL_0	✓	✓	✓	✓	✓
14	TTL 1	KY_TTL_1	✓	✓	✓	✓	✓
15	TTL 2	KY_TTL_2	✓	✓	✓	✓	✓
16	TTL 3	KY_TTL_3	✓	✓	✓	✓	✓
17	TTL 4	KY_TTL_4	✓	✓	✓	✓	✓
18	TTL 5	KY_TTL_5	✓	✓	✓	✓	✓
19	TTL 6	KY_TTL_6	✓	✓	✓	✓	✓
20	TTL 7	KY_TTL_7	✓	✓	✓	✓	✓
21	LVTTL 0	KY_LVTTL_0	✓	✓	✓	✓	✓
22	LVTTL 1	KY_LVTTL_1	✓	✓	✓	✓	✓
23	LVTTL 2	KY_LVTTL_2	✓	✓	✓	✓	✓
24	LVTTL 3	KY_LVTTL_3	✓	✓	✓	✓	✓
25	LVTTL 4	KY_LVTTL_4	✓	✓	✓	✓	✓
26	LVTTL 5	KY_LVTTL_5	✓	✓	✓	✓	✓
27	LVTTL 6	KY_LVTTL_6	✓	✓	✓	✓	✓
28	LVTTL 7	KY_LVTTL_7	✓	✓	✓	✓	✓
29	OptoCoupled Output 0						
30	OptoCoupled Output 1						
31	OptoCoupled Output 2						
32	OptoCoupled Output 3						
33	OptoCoupled Output 4						
34	OptoCoupled Output 5						
35	OptoCoupled Output 6						
36	OptoCoupled Output 7						
37	LVDS Output 0						
38	LVDS Output 1						
39	LVDS Output 2						
40	LVDS Output 3						
41	Camera Trigger	KY_CAM_TRIG			✓		
42	Continuous	KY_CONTINUOUS		✓			
43	Software	KY_SOFTWARE		✓	✓		✓
44	Encoder 0	KY_ENCODER_0		✓	✓		✓
45	Encoder 1	KY_ENCODER_1		✓	✓		✓
46	Encoder 2	KY_ENCODER_2		✓	✓		✓
47	Encoder 3	KY_ENCODER_3		✓	✓		✓
48	Timer 0 Active	KY_TIMER_ACTIVE_0	✓	✓	✓		✓
49	Timer 1 Active	KY_TIMER_ACTIVE_1	✓	✓	✓		✓
50	Timer 2 Active	KY_TIMER_ACTIVE_2	✓	✓	✓		✓
51	Timer 3 Active	KY_TIMER_ACTIVE_3	✓	✓	✓		✓
52	Timer 4 Active	KY_TIMER_ACTIVE_4	✓	✓	✓		✓
53	Timer 5 Active	KY_TIMER_ACTIVE_5	✓	✓	✓		✓
54	Timer 6 Active	KY_TIMER_ACTIVE_6	✓	✓	✓		✓
55	Timer 7 Active	KY_TIMER_ACTIVE_7	✓	✓	✓		✓
56	User Output 0	KY_USER_OUT_0	✓				

57	User Output 1	KY_USER_OUT_1	✓				
58	User Output 2	KY_USER_OUT_2	✓				
59	User Output 3	KY_USER_OUT_3	✓				
60	User Output 4	KY_USER_OUT_4	✓				
61	User Output 5	KY_USER_OUT_5	✓				
62	User Output 6	KY_USER_OUT_6	✓				
63	User Output 7	KY_USER_OUT_7	✓				
64	TriggerN 0	TriggerN0	✓	✓	✓	✓	✓
65	TriggerN 1	TriggerN1	✓	✓	✓	✓	✓
66	TriggerN 2	TriggerN2	✓	✓	✓	✓	✓
67	TriggerN 3	TriggerN3	✓	✓	✓	✓	✓
68	TriggerN 4	TriggerN4	✓	✓	✓	✓	✓
69	TriggerN 5	TriggerN5	✓	✓	✓	✓	✓
70	TriggerN 6	TriggerN6	✓	✓	✓	✓	✓
71	TriggerN 7	TriggerN7	✓	✓	✓	✓	✓
72	TriggerN 8	TriggerN8	✓	✓	✓	✓	✓
73	TriggerN 9	TriggerN9	✓	✓	✓	✓	✓
74	TriggerN 10	TriggerN10	✓	✓	✓	✓	✓
75	TriggerN 11	TriggerN11	✓	✓	✓	✓	✓
76	TriggerN 12	TriggerN12	✓	✓	✓	✓	✓
77	TriggerN 13	TriggerN13	✓	✓	✓	✓	✓
78	TriggerN 14	TriggerN14	✓	✓	✓	✓	✓
79	TriggerN 15	TriggerN15	✓	✓	✓	✓	✓

Table 14 – Frame Grabber I/O source

4.7 Pulse Message

According to CLHS standard, the Pulse Message's primary use is to cause the camera to perform exposure and send out the video frame(s). KAYA's CLHS compatible Frame Grabber implements the Pulse Message interface for triggering camera's frame acquisition, subject to firmware and software capabilities.

The software model provides two modes of operation, a basic one which can be used to send a Mode1 message upon trigger reception and an advanced one that enables to configure the message that will be sent out. In order to use the extended Pulse Message interface, the "PulseMessageMode" Frame Grabber parameter should be changed to "Advanced". Trigger source, characteristics and Pulse Message fields should be configured in order to generate the required Pulse Message.

Pulse Message fields' description can be found in CLHS official document under the "Pulse Message" section.
The Frame Grabber triggering system description can be found in section 5.2. The Pulse Message control parameters are described in the table below:

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: ExtendedStreamFeatures \ PulseMessageSelector						
Pulse Message Mode	Pulse Message operation mode	PulseMessageMode	Enumeration	0 1	Basic Advanced	Select "Advanced" to activate these features. In "Basic" mode, Pulse Message Mode1 messages will be issued.
Pulse Message Selector	Selected Pulse Message configuration	PulseMessageSelector	Enumeration	0 1 2 3 4 5 6 7	PulseMessage0 PulseMessage1 PulseMessage2 PulseMessage3 PulseMessage4 PulseMessage5 PulseMessage6 PulseMessage7	
Pulse Message Enable	Controls if the trigger is active	PulseMessageEnable [PulseMessageSelector]	Enumeration	0 1	Off On	
Pulse Message Activation	Activation mode of the trigger in respect to the input.	PulseMessageActivation [PulseMessageSelector]	Enumeration	0 1 2 3 4 5	RisingEdge FallingEdge AnyEdge RisingEdgeInv FallingEdgeInv AnyEdgeInv	Inv means inverted. Only Selected edge packet will be issued to the camera.
Pulse Message Source	Source I/O	PulseMessageSource [PulseMessageSelector]	Enumeration			See section Link Trigger Source options
Pulse Message Delay	Delay before issuing trigger	PulseMessageDelay [PulseMessageSelector]	Integer			In units of microseconds (us)
Pulse Message Filter	Filter for frame grabber trigger	PulseMessageFilter [PulseMessageSelector]	Float			In units of microseconds (us) 8ns resolution using fraction value
Pulse Message Software	Generates an internal trigger	PulseMessageSoftware [PulseMessageSelector]	Command	1 - Activate		To issue command "PulseMessageSource" must be set to "Software"

Pulse Message Link Mask Enable	The output physical link mask	PulseMessageLinkMaskEnable [PulseMessageSelector]	Integer			
Pulse Message Pulse Mode	Pulse Mode of generated Pulse Message	PulseMessagePulseMode [PulseMessageSelector]	Enumeration	1 2 3 4 5 6 7	Mode1 Mode2 Mode3 Mode4 Mode5 Mode6 Mode7	According to CLHS Pulse Message specification
Pulse Message Color	Pulse message color select definition	PulseMessageColor [PulseMessageSelector]	Enumeration	0 1 2 3	All Red Blue Green	According to CLHS Pulse Message specification
Pulse Message Pulse Effect	Pulse Message effect intended for synchronous frame by frame control of camera features	PulseMessagePulseEffect [PulseMessageSelector]	Integer			According to CLHS Pulse Message specification
Pulse Message Synchronization Request	Request for reset to an initial frame or line count and agree on the identification of the image data for parallel processing applications	PulseMessageSyncRequest [PulseMessageSelector]	Boolean			According to CLHS Pulse Message specification
Pulse Message Frame Period	Pulse Message frame period down counter definition	PulseMessageFramePeriod [PulseMessageSelector]	Integer			According to CLHS Pulse Message specification
Pulse Message Integration Start Red	Pulse Message Red channel integration period down counter definition	PulseMessageIntegrationStartRed [PulseMessageSelector]	Integer			According to CLHS Pulse Message specification
Pulse Message Integration Start Green	Pulse Message Green channel integration period down counter definition	PulseMessageIntegrationStartGreen [PulseMessageSelector]	Integer			According to CLHS Pulse Message specification
Pulse Message Integration Start Blue	Pulse Message Blue channel integration period down counter definition	PulseMessageIntegrationStartBlue [PulseMessageSelector]	Integer			According to CLHS Pulse Message specification

Table 15 – Pulse Message control parameters

4.7.1 Timer Control

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: I/O Control/Timer Control						
Timer Selector	Selects the trigger to configure	TimerSelector	Enumeration (Selector)			See Table 17**Error! Reference source not found.**
Timer Delay	Sets the duration in microseconds(us) of the delay to apply at the reception of a trigger before starting the Timer	TimerDelay [TimerSelector]	Float	0.016000 – 34359738		In units of microseconds (us)
Timer Duration	Sets the duration in microseconds(us) of the Timer pulse	TimerDuration [TimerSelector]	Float			In units of microseconds (us)
Timer Activation	Specifies the activation mode of the trigger	TimerActivation [TimerSelector]	Enumeration	0 1 2 6 7	RisingEdge FallingEdge AnyEdge LevelHigh LevelLow	
Timer Output Inverter	Controls the inversion of the timer output signal	TimerOutputInverter [TimerSelector]	Boolean	0 1	False True	
Timer Trigger Source	Selects the source of the trigger to start the Timer	TimerTriggerSource [TimerSelector]	Enumeration			See section Link Trigger Source options
Timer Trigger Software	Generates an internal trigger. Timer trigger source must be set to Software	TimerTriggerSoftware [TimerSelector]	Command	1 - Activate		
Timer Reset	Does a software reset of the selected timer and starts it. The timer starts immediately after the reset unless a timer trigger is active	TimerReset [TimerSelector]	Command	1 - Activate		
Timer Event Mode	Selects the activation mode of the trigger	TimerEventMode [TimerSelector]	Enumeration	0 2 3 4	Disabled RisingEdge FallingEdge AnyEdge	

Table 16 – Timer Control parameters

Value	Output	Gen<i>Cam parameter name
0	Timer Active 0	Timer0
1	Timer Active 1	Timer1
2	Timer Active 2	Timer2
3	Timer Active 3	Timer3
4	Timer Active 4	Timer4
5	Timer Active 5	Timer5
6	Timer Active 6	Timer6
7	Timer Active 7	Timer7

Table 17 – Timer selection options

Configure am internal timer for timed trigger generation. Incorporate selection of signal edge capture mode, timer signal delay and duration and inverter for timer signal.

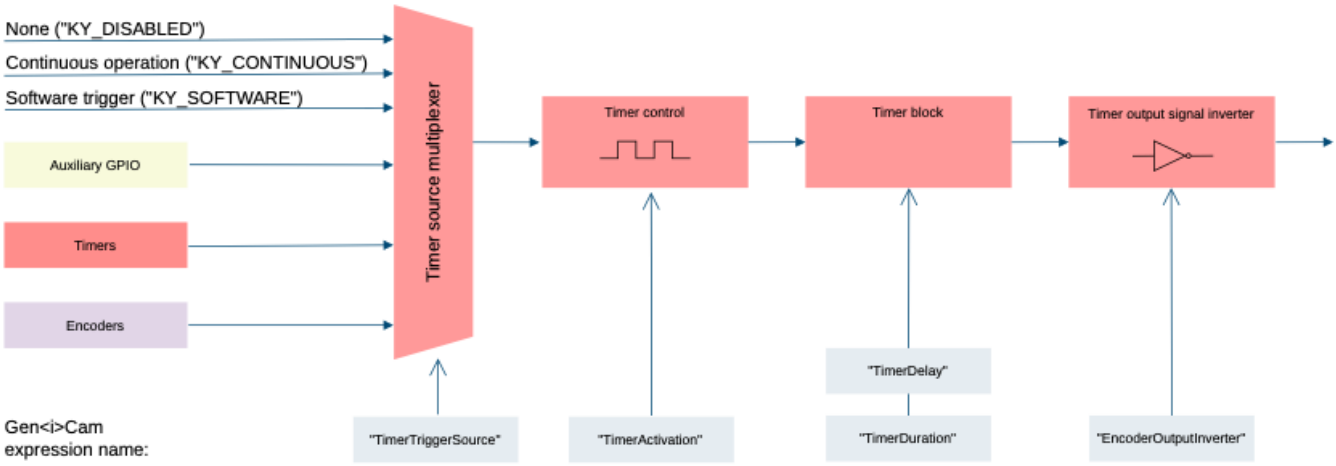


Figure 4 – Timer triggers structure

4.7.1.1 Timer activation mode

The trigger activation mode configures the capture criteria of signal state. Default value is Rising Edge, which will issue a trigger on signal rising edge event. The different modes functionality is as follows:

1. Any Edge: Any edge of the selected trigger source signal will increment 1 timer count (Duration + Delay time).
2. Rising Edge: A rising edge of the selected trigger source will increment 1 timer count (Duration + Delay time), and a falling edge is ignored.
3. Falling Edge: A falling edge of the selected trigger source will increment 1 timer count (Duration + Delay time), and a rising edge is ignored.
4. Level High: High signal level enables a continuous timer operation. Low signal level will halt the timer.
5. Level Low: Low signal level enables a continuous timer operation. High signal level will halt the timer.

4.7.1.2 Timer delay, duration and signal inversion

Input value of delay, duration and inversion will determine the timer signal behavior as a rule for timer tick count. Duration will determine the ON position of the timer signal, while delay will determine the OFF position of the signal. The output inverter is responsible for flipping the signal level of duration and delay values.

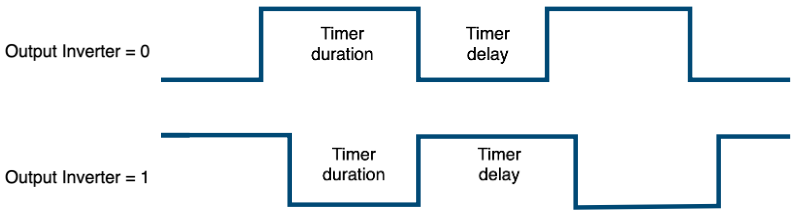


Figure 5 – Output inverters

Timers' counters are in 8ns intervals for high resolution functionality. Available interface in API provides input in microsecond; nevertheless, to achieve higher resolution, relevant fraction values should be entered after the decimal point.

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4.7.1.3 Timer Event

Timer trigger event may be enabled for selected timer. This will generate event callback whenever such trigger is generated in hardware. Steps to enable and use such event mechanism are as follows:

1. Select timer by setting the “TimerSelector” parameter and select signal capture mode using “TimerEventMode” for selected timer.
2. Register a callback function for Auxiliary events using KYVPLibExtension library KYVPExtension_PCIInterface_AuxDataCallback_Register() function.
3. To access the data attached to such event please refer to KYVP_AUX_DATA pointer in the callback.

4.7.2 TriggerN Control

According to CoaXPress v2.0 standard some cameras can send triggers to the Frame Grabber in CoaXPress packets. These triggers are received by Frame Grabber and forwarded to other components.

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: FrameGrabberIOControl \ TriggerNControl						
TriggerN Count	Maximum number of available TriggerN configurations	TriggerNCount	Integer			
TriggerN Selector	Select the TriggerN configuration to be modified	TriggerNSelector	Enumeration (Selector)			See Table 19**Error! Reference source not found.**
TriggerN Mode	Activate/deactivate the selected TriggerN configuration.	TriggerNMode [TriggerNSelector]	Enumeration	0 1	Off On	
TriggerN Camera Source	Source camera index which the LinkTriggerN trigger will originate from	TriggerNCameraSource [TriggerNSelector]	Integer			
TriggerN Source	The trigger number to pass through by this configuration	TriggerNSource [TriggerNSelector]	Integer	Min: 0 Max: 15		
TriggerN Event Mode	TriggerN Event Mode enable	TriggerNEventMode	Enumeration	0 1	Off On	

Table 18 – Available configurations for TriggerN Control

4.7.2.1 TriggerN mode

Activate/deactivate the selected TriggerN configuration. If active it will pass through only the LinkTriggerN trigger selected by "TriggerNCameraSource" and "TriggerNSource".

4.7.2.2 TriggerN Source

The trigger number, defined by the LinkTriggerN field in the source trigger packet, to pass through by this configuration.

Value	TriggerN	Gen<i>Cam parameter name
0	TriggerN 0	TriggerN0
1	TriggerN 1	TriggerN1
2	TriggerN 2	TriggerN2
3	TriggerN 3	TriggerN3
4	TriggerN 4	TriggerN4
5	TriggerN 5	TriggerN5
6	TriggerN 6	TriggerN6
7	TriggerN 7	TriggerN7
8	TriggerN 8	TriggerN8
9	TriggerN 9	TriggerN9
10	TriggerN 10	TriggerN10
11	TriggerN 11	TriggerN11
12	TriggerN 12	TriggerN12
13	TriggerN 13	TriggerN13
14	TriggerN 14	TriggerN14
15	TriggerN 15	TriggerN15

Table 19 – TriggerN selector

How to configure TriggerN control:
1. Identify the camera index and the high speed trigger number that need to be captured by the Frame Grabber.
2. Select one of the available TriggerN configuration using "TriggerNSelector".
3. Activate the configuration to allow the high speed trigger to pass through using "TriggerNMode" = "On".
4. Set the camera index and the high speed trigger number, defined by the LinkTriggerN field in the source trigger packet, using "TriggerNCameraSource" and "TriggerNSource" respectively.

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4.8 CoaXPress

Configuration parameters of Connection Test, using dedicated test pattern packets produced by a sequence generator. As described in JIIA CXP-001-2013 (CoaXPress Standard) document section 8.7 - “Connection Test”.

The trigger statistics track output and input trigger signals. The relevant counters increase according to how the trigger signals influence the system.

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: CoaXPress						
CXP Heartbeats Throttling	CXP Heartbeats Throttling	CxpHeartbeatsThrottling [CxpConnectionSelector]	Integer			
Gen<i>Cam Category: CoaXPress/CoaXPress Connection Selector						
CoaXPress Connection Selector	Selects the CoaXPress physical connection to control	CxpConnectionSelector	Integer (Selector)			FG_MAX_VALUE-1
Connection Test Mode	Test communication errors of the system cabling between devices	CxpConnectionTestMode [CxpConnectionSelector]	Enumeration	0 1	Off Mode1	Mode1 will enable traffic of connection test packets from Host to Device
Connection Test Error Count	Camera CRC Error Counter. Number of CRC errors generated from corrupted data packets	CxpConnectionTestErrorCount [CxpConnectionSelector]	Integer			
Connection Test RX packets	Reports the current count for test packets received by the device on the connection selected by CxpConnectionSelector	CxpConnectionTestRxPacketCount [CxpConnectionSelector]	Integer (8 bytes)			
Connection Test TX packets	Reports the current count for test packets sent to the device on the connection selected by CxpConnectionSelector	CxpConnectionTestTxPacketCount [CxpConnectionSelector]	Integer (8 bytes)			
Connection Test Counters Reset	Reset all connection test counters	CxpConnectionTestCountersReset	Command	1 - Activate		
Trigger Missed Count	Missed triggers count, this counter increases if\when a trigger arrives from the user	TriggerMissedCount [CxpConnectionSelector]	Integer			
Trigger Sent Count	Sent triggers count, this counter increases if\when a trigger packet is sent from the host IP to remote device	TriggerSentCount [CxpConnectionSelector]	Integer			
Trigger Acknowledge Count	Acknowledgement triggers count, this counter increases if\when an acknowledge arrives from the remote device to the host, for a trigger sent from the host to the remote device	TriggerAcknowledgeCount [CxpConnectionSelector]	Integer			

Trigger Change Count	In change triggers count, this counter increases if\when the user gives a trigger to the host	TriggerChangeCount [CxpConnectionSelector]	Integer			
Trigger Counters Reset	Reset all trigger counters	TriggerCountersReset	Command	1 - Activate		

Table 20 – CoaXPress parameters

4.9 System Monitor Control

Monitor and configuration of available individual system components, independent of currently connected peripherals. For example: stream channel controls, link connection configurations and monitoring etc.

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: System Monitor Control / Connection Channel Selector						
Connection Channel Selector	Selects the connection channel to control	ConnectionChannelSelector	Integer (Selector)			
Connection Speed	Connection speed change	ConnectionSpeed [ConnectionChannelSelector]	Enumeration	0x28	ConnectionSpeedCXP1	
				0x30	ConnectionSpeedCXP2	
				0x38	ConnectionSpeedCXP3	
				0x40	ConnectionSpeedCXP5	
				0x48	ConnectionSpeedCXP6	
				0x50	ConnectionSpeedCXP10	
				0x58	ConnectionSpeedCXP12	
				10	ConnectionSpeedCLHS10	
				12	ConnectionSpeedCLHS12	
				14	ConnectionSpeedCLHS14	
				16	ConnectionSpeedCLHS16	
Revision Packets	Number of Revision packets	ConnectionRevisionPackets [ConnectionChannelSelector]	Integer			
Command Packets	Number of Command packets	ConnectionCommandPackets [ConnectionChannelSelector]	Integer			
FEC Received Packets	Received packets by FEC	ConnectionFECRxPackets [ConnectionChannelSelector]	Integer			
FEC Corrected Packets	Number of corrected packets by FEC	ConnectionFECCorrectedPackets [ConnectionChannelSelector]	Integer			
FEC Corrupted Packets	Number of uncorrectable packets	ConnectionFECCorruptedPackets [ConnectionChannelSelector]	Integer			
PCS Test Mode	PCS Test Mode	PCS_TestMode [ConnectionChannelSelector]	Enumeration	0x00000000	PCSTEST_NORMAL	
				0x00000001	PCSTEST_PRBS7	
				0x00000002	PCSTEST_PRBS9	
				0x00000003	PCSTEST_PRBS15	
				0x00000004	PCSTEST_PRBS23	
				0x00000005	PCSTEST_PRBS31	
				0x00000006	PCSTEST_HIGHFREQDATA	
				0x00000007	PCSTEST_LOWFREQDATA	
XGMII Test Mode	XGMII Test Mode	XGMII_TestMode [ConnectionChannelSelector]	Enumeration	0x00000000	XGMIITEST_NORMAL	
				0x00000001	XGMIITEST_LOOPBACK	
				0x00000002	XGMIITEST_PATTERNGENERATOR	
XGMII Tester TX Packets	XGMII tester transmitted packets	XGMIITester_TXPackets [ConnectionChannelSelector]	Integer			
XGMII Tester RX Packets	XGMII tester received packets	XGMIITester_RXPackets [ConnectionChannelSelector]	Integer			

XGMII Tester Error Packets	XGMII tester error packets	XGMIITester_ErrorPackets [ConnectionChannelSelector]	Integer		
Packets Connection Counters Reset	Reset all connection counters	ConnectionStatisticsReset [ConnectionChannelSelector]	Command		
Gen<i>Cam Category: System Monitor Control / Stream Channel Selector					
Stream Channel Selector	Selects the stream channel to control	StreamStatisticsReset [StreamChannelSelector]	Integer		
Statistics Counters Reset	Reset all stream statistics counters	StreamStatisticsReset [StreamChannelSelector]	Command	1 - Activate	
CRC Error Counter	Camera CRC Error Counter. Number of CRC errors generated from corrupted data packets	StreamCRCErrorCounter [CameraSelector]	Integer		Errors are generated from corrupted data packets
RX Packet Counter	Stream RX Packet Counter. Total number of packets received from the camera	StreamRxPackets [StreamChannelSelector]	Integer		
Drop Packet Counter	Camera Drop Packet Counter. Number of packets dropped due to corruption	StreamDropPackets [StreamChannelSelector]	Integer		
RX Frame Counter	RX Frame Counter. Number of received frames from camera	StreamRxFrames [StreamChannelSelector]	Integer		
Drop Frame Counter	Camera Drop Frame Counter. Number of frames dropped due to corruption	StreamDropFrames [StreamChannelSelector]	Integer		
Drop StreamId Counter	Camera Drop Stream Id Counter. Number of frames dropped due to StreamId corruption	StreamDropStreamIdCounter	Integer		
RX Line Counter	RX Line Counter	StreamRXLineCounterReg	Integer		
RX IH Counter	RX Image Header Counter	StreamRXIHCounter	Integer		
Drop IH Counter	Drop Image Header Counter	StreamDropIHCounter	Integer		

Table 21 – System Monitor Control parameters

4.10 Transport Channel Selector

KAYA's Frame Grabber implements the Transport Channel Selector. It indicates the low level connection packets sent between the Host and Device, used for link synchronization. In case of missing or unstable connection, the counters will indicate attempts to resynchronize the link.

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam	
Gen<i>Cam Category: StreamFeatures \ TransportLayerControl						
Transport Channel Selector	Selects the transport channel of the stream for monitoring and control	TransportChannelSelector	Integer			
Revision Packets	Number of Revision packets on master link of the device	RevisionPackets [CameraSelector]	Integer			
Command Packets	Number of Command packets on master link of the device	CommandPackets [CameraSelector]	Integer			
FEC Received Packets	Received packets by FEC on master link of the device	FEC_RXPackets [CameraSelector]	Integer			
FEC Corrected Packets	Number of corrected packets by FEC on master link of the device	FEC_CorrectedPackets [CameraSelector]	Integer			
FEC Corrupted Packets	Number of uncorrectable packets on master link of the device	FEC_CorruptedPackets [CameraSelector]	Integer			
Transport Counters Reset	Rest all transport counters	TransportCountersReset	Command	1 - Activate		

Table 22 –Transport Channel Selector

4.11 Manual Detection Configuration

Parameters are displayed dynamically according to the current configuration. Selecting a parameter value can enable additional configuration parameters required by the selected setting.

After a manual detection configured, initiate the scanning process using the KYVPLibTL library KYVPLibTL_IFUpdateDeviceList() function. The camera discovery parameters are described in the following table.

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam	
Gen<i>Cam Category: Manual Detection Configuration						
No Devices Access	If enabled, no operations requiring access to remote device will be performed	NoDevicesAccess	Boolean			
Suppress Link ID Verification	If enabled, link ID verification for the configured device will be suppressed. This will allow devices to be detected even if the configuration does not reference the device's correct link	SuppressLinkIDVerification	Boolean			
Device Count	Number of devices selected for manual detection	ManualDetectionDevicesCount	Integer			
Link Count	Select how many camera links are active for Manual Detection Device 0	ManualDetectionDevice_0_LinksCount	Enumeration	LINKS_1 LINKS_2 LINKS_4 LINKS_8	1 link 2 links 4 links 8 links	See Note [1]
Link Speed	Sets the link speed in Gbps for all active links of Manual Detection Device 0	ManualDetectionDevice_0_LinksSpeed	Enumeration	SPEED_1_25G SPEED_2_5G SPEED_3_125G SPEED_5G SPEED_6_25G SPEED_10G SPEED_12G	1.25 Gbps 2.5 Gbps 3.125 Gbps 5 Gbps 6.25 Gbps 10 Gbps 12 Gbps	See Note [1]
Link 0	Map camera Link 0 to a specific host link, or leaves it unassigned	Manual DetectionDevice_0_Link 0	Enumeration	HOST_LINK_NOT_ASSIGNED HOST_LINK_0 HOST_LINK_1 HOST_LINK_2 HOST_LINK_3	Not assigned Host Link 0 Host Link 1 Host Link 2 Host Link 3	See Note [1]

Table 23 – Device Control parameters

NOTE:

1. The table above shows the parameters for one link 0 as an example. If multiple links are configured, an identical set of parameters is displayed for each link, supporting up to eight links.

4.11.1 Manual camera discovery

Manual camera discovery should be performed with the presumption that camera connectivity topology and communication speed is known for current discovery session. Generally, Manual discovery is much faster and less restrictive. This procedure allows the user to specify known camera parameters instead of retrieving them from a connected camera. It can be helpful in cases:

- Debugging when automatic detection fails.
- The camera does not implement read/write functionality over a connection with grabber.
- we need to skip the camera reset/initialization process.

Nevertheless, **wrong Manual connectivity configurations might yield in unknown results and insufficient camera initialization**.

Manual discovery process steps:

1. Determine the camera speed, number of links and order of connection between camera channels and Frame Grabber links. All the camera parameters should correspond to the current connection topology of the camera.
2. Set the ManualDetectionDeviceCount parameter to desired number of detected cameras.
3. Set the links number and connection speed for each camera.
4. For each physical connection select each camera link to which the Frame Grabber channel is attached.
5. Now the camera scan can be initiated using the KYVPLibTL_IFUpdateDeviceList() function.

An example of how to determine these values is provided below, using a 4-link camera to the 4links Frame Grabber connection:

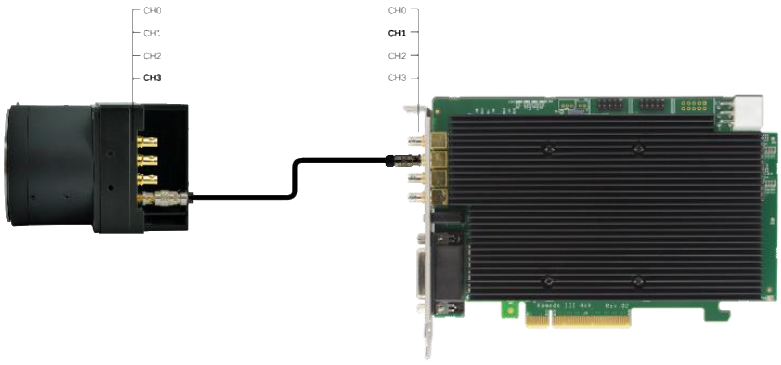


Figure 6 – An example of device connection for Manual Camera Discovery using

Camera parameters:

- Device 1
- Link 1

Connection parameters:

- Speed 3.125
- Host Link 1
- Device Link 3

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5 Local Device Control

The Local Device serves as a proxy for one physical remote device on the PCI Interface side, enabling communication and managing Data Stream modules. Local Device is a certain set of parameters that are actually on the grabber's side but logically relate to a camera. Adjusting its parameters affects camera parameters that are actually related to the camera, but are on the Frame Grabber side, change.

5.1 Device Control

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam	
Gen<i>Cam Category: Device control						
Connection Mask	Bitmask indicating grabber links on which camera is detected	LinksConnectionMask	String			

Table 24 – Device Control parameters

5.2 Camera Trigger Control

The Camera triggers are issued per camera through the camera CoaXPress channels. Camera logic intercepts the signal and performs according to preconfigured camera setting, such as 1 frame transmission for example. A sequence of synchronous or asynchronous signals can be configured to be issued for selected camera. Such configuration can be useful in configuring event-controlled image acquisition.
The flow of the camera trigger signal can be seen in Figure below:

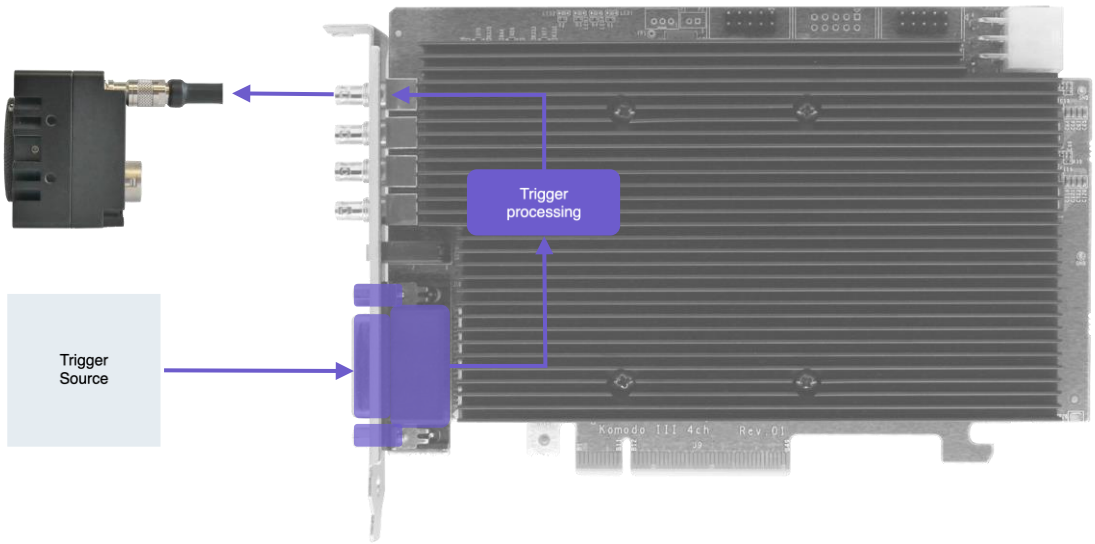


Figure 7 – Camera trigger source

Triggers' origin can be selected from number of sources such as encoders, I/O lines and timers. Additional properties are available for better capturing and processing trigger signals (To configure camera trigger mode please refer to the camera manufacturer manual). The structure of the camera trigger is described in figure below:

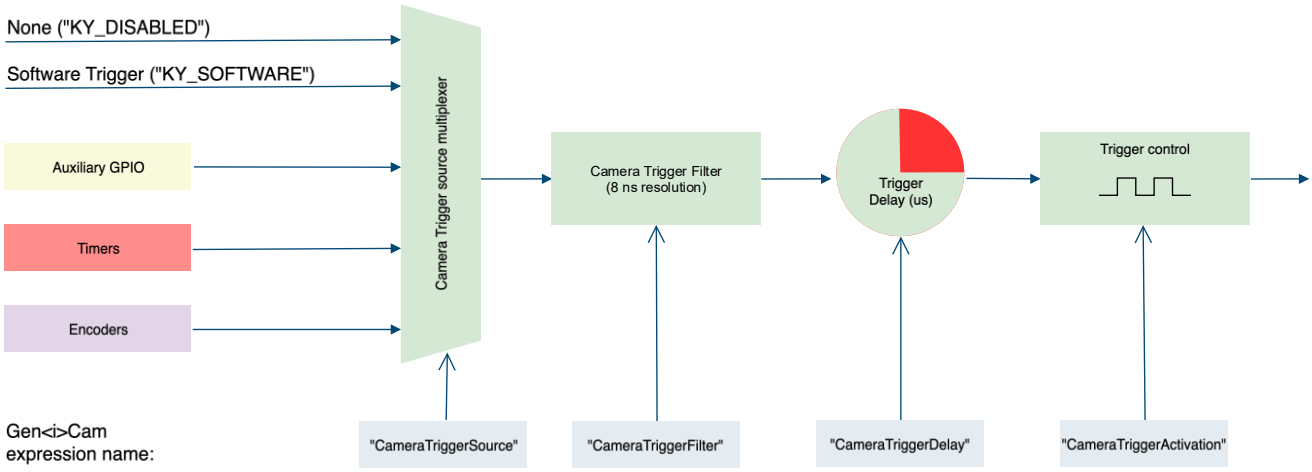


Figure 8 – Camera trigger structure

The parameters of the camera trigger are described in the table below:

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: ExtendedStreamFeatures \ CameraTriggerControl						
Camera Trigger Mode	Controls if the trigger is active	CameraTriggerMode [CameraSelector]	Enumeration	0 1	Off On	
Camera Trigger Activation	Activation mode of the trigger in respect to the input	CameraTriggerActivation [CameraSelector]	Enumeration	0 1 2 3 4 5	RisingEdge FallingEdge AnyEdge RisingEdgeInv FallingEdgeInv AnyEdgeInv	Inv means inverted. Only Selected edge CXP packet will be issued to the camera.
Camera Trigger Source	Source I/O	CameraTriggerSource [CameraSelector]	Enumeration			See section Link Trigger Source options
Camera Trigger Delay	Delay before issuing trigger	CameraTriggerDelay [CameraSelector]	Integer			In units of microseconds (us)
Camera Trigger Filter	Filter for Frame Grabber trigger	CameraTriggerFilter [CameraSelector]	Float			In units of microseconds (us) 8ns resolution using fraction value
Camera Trigger Software	Generates an internal trigger	CameraTriggerSoftware [CameraSelector]	Command	1 – Activate		To issue command “CameraTriggerSource ” must be set to “Software”
Camera Trigger Event Enable	Enables event generation for camera trigger	CameraTriggerEventEnable [CameraSelector]	Enumeration	0 1	Disable Enable	Will generate software even for any trigger

Table 25 – Camera Trigger parameters

5.2.1 Camera Trigger activation mode

The trigger activation mode configures the capture criteria of signal state. Default value is Rising Edge, which will issue a trigger on signal rising edge event. The different modes functionality is as follows:

1. Any Edge: A rising edge of the selected trigger source generates rising edge trigger packets, and a falling edge generates falling edge packets. This allows e.g. camera exposure to be controlled by the time between the rising and falling edges, as well as one of the edges providing the trigger.
2. Rising Edge: A rising edge of the selected trigger source generates rising edge trigger packets, and a falling edge is ignored. This allows a higher trigger rate but does not allow exposure control independent from the trigger rate.
3. Falling Edge: A falling edge of the selected trigger source generates falling edge trigger packets, and a rising edge is ignored.
4. Inverted: This mode can be applied to any of the above, and results in a rising edge generating a falling edge trigger packet, and a falling edge to generate a rising edge trigger packet.

5.2.2 Camera Trigger Delay

The trigger delay is a mechanism for postponing the incoming signal for a specified number of microseconds. As a result, trigger will be issued after specified time delay to overcome known system latency. To disable, value 0 should be set.

5.2.3 Camera Trigger signals filter

The filter of the trigger signals acts as a de-bouncing mechanism for better handling generated noise. By default, the filter is disabled with the value of 0. The signal filter resolution can be set at 8ns intervals for high resolution functionality. If the trigger filter is set to a larger value than the width of the trigger pulse, then the pulse will be filtered out and no trigger will occur. Available interface in API provides input in microsecond; nevertheless, to achieve higher resolution, relevant fraction values should be entered after the decimal point.

5.2.4 Camera Trigger Event Enable

Camera trigger event may be enabled for selected camera. This will generate event callback whenever such trigger is generated in hardware. Steps to enable and use such event mechanism are as follows:

1. "CameraTriggerEventEnable" is a Local Device parameter subordinate to connected camera. Use the KYVPParametersHandler library function KYParametersHandler_SetValueEnumByValueName() to set the parameter value to "On".
2. Register a callback function for Auxiliary events using KYVPLibExtension library KYVPExtension_PCIInterface_AuxDataCallback_Register() function.
3. To access the data attached to such event please refer to KYVP_AUX_DATA pointer in the callback.

5.2.5 Steps to properly configure Camera Triggers

1. "CameraTriggerMode" is a grabber parameter subordinate to connected camera. Use the KYVPParametersHandler library function KYParametersHandler_SetValueEnumByValueName() to set the parameter value to "On".
2. The trigger source should be selected according to provided sources and available card GPIO. Only one source can be active, for each camera, at any time.
3. The Trigger Filter resolution ("CameraTriggerDelay"), Activation Mode ("CameraTriggerActivation") and Trigger Delay ("CameraTriggerDelay") parameters should be configured according to desired output.
4. In some cases, the trigger sources should also be configured via provided API before trigger configuration is complete. (e.g if "KY_TIMER_ACTIVE_0" is to be selected as Camera Trigger source, then "Timer0" should first be configured as described in Timer Block configuration in this chapter).
5. Configure the camera to be in trigger mode, in order to allow triggered control through the Frame Grabber. NOTE*: To configure camera trigger mode please refer to camera manufacturer manual.*

6 Stream Control

The Stream represents a single image data stream from a Remote Device, serving as the acquisition engine and managing the internal buffer pool. It also offers signaling and configuration options. A device can support zero, one or multiple data streams, it limited only by the hardware and implementation.
The Stream configurations contain various features and controls such as image format, stream triggers, stream statistics etc.

6.1 Image Format Control

The image format control is responsible for configuring some of the image processing features. The image format control can be found in the following table.

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: ImageFormatControl						
Segments per Buffer	Number of Lines/Frames to accumulate in a single buffer frame	SegmentsPerBuffer [CameraSelector]	Integer	≥ 1		This feature is mostly used for LineScan cameras. Please refer to 6.1.1 for more details.
Transformation Pixel Format	Pixel Format for Image Format transformation. Availability is subject to current camera's pixel format	PixelFormat [CameraSelector]	Enumeration	0000 0101 0102 0103 0104 0105 0401 0402 0403 0404 0405 0501 0311 0321 0331 0341	Normal Mono8 Mono10 Mono12 Mono14 Mono16 RGB8 RGB10 RGB12 RGB14 RGB16 RGBA8 BayerGR8 BayerRG8 BayerGB8 BayerBG8	Conversion is possible according to input camera PixelFormat, resolution and HW capabilities
Debayer Mode	Bayer demosaic algorithm	DebayerMode	Enumeration	Demosaic3x3 Demosaic3x2	Demosaic 3x3 Demosaic 3x2	
Width	Width of the image, to override the one provided by the remote device (in pixels)	Width [CameraSelector]	Integer			
Height	Height of the image, to override the one provided by the remote device (in pixels)	Height [CameraSelector]	Integer			
Packed Data Mode	Select maintain data packing mode for stream output	PackedDataMode	Enumeration	0000 0001 0003	Unpacked Packed_RowAligned32 Packed_RowAligned32_Reverse	

Table 26 – Image Format control parameters

NOTES:

1. A “Demosaic3x3” is a 3x3 hardware debayer that almost doesn't add any processing latency, 3x2 hardware debayer should be used for LineScan camera with 2 rows.
2. The captured images will be as bayer format raw data. Capturing images operation can be executed using API functions. More information about API functions can be found in Vision_Point_II_API_Data_Book.

6.1.1 Segment per buffer

Stream configuration to capture several frames/lines before an event signal is received in software. This feature is mostly used for LineScan cameras – several lines are accumulated before software receives indication signal on new data acquisition. This prevents the software from receiving lines too frequently thus relieving the CPU operation.
“SegmentsPerBuffer” parameter should be set using KYParametersHandler_SetValue(). By default, “SegmentsPerBuffer” value is 1 which means that software indication signal will occur on every frame/line captured.

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6.1.2 Bayer demosaic

A Bayer filter mosaic is a color filter array (CFA) for arranging RGB color filters on a square grid of photo sensors. Its particular arrangement of color filters is used in most single-chip digital image sensors used in digital cameras, camcorders, and scanners to create a color image. The Bayer filter has twice green pixels then red or blue ones because human's eye is more sensitive to green light. The filter pattern is 50% green, 25% red and 25% blue, hence is also called RGBG, GRGB, BGGR or RGGB. The example structure of the CFA can be seen in Figure 9.

As each pixel in the array contains only one color plane, the demosaicking algorithm should calculate the missing color pixels at each particular position. To enable the demosaic format transformation, set the "PixelFormat" parameter value to "RGB8". **Note that buffer allocation is directly affected by this operation!**

Two different demosaicking algorithms are available dependent on the line scan or area scan sensor.

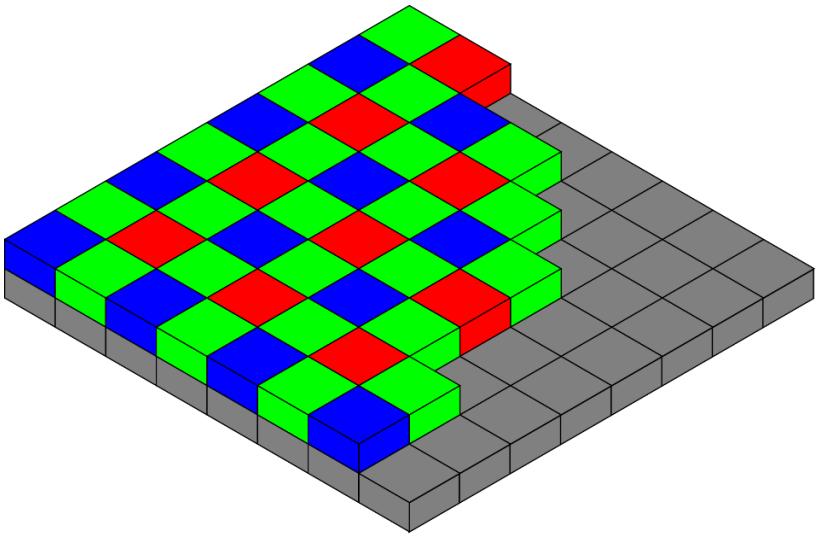


Figure 9 – Bayer filter example

6.1.2.1 Bilinear demosaicking (Area scan)

The bilinear demosaicking algorithm performs the color reconstruction for each pixel by interpolation in a 3-by-3 pixel neighborhood. The interpolation kernel differs for even/odd rows/columns and is according to the Figures shown below. The calculations are performed with full 16bits resolution.

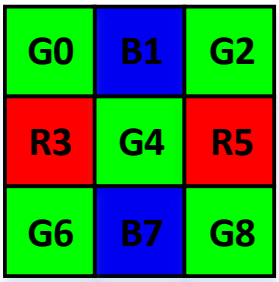
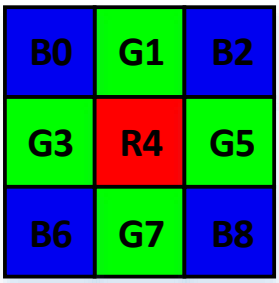
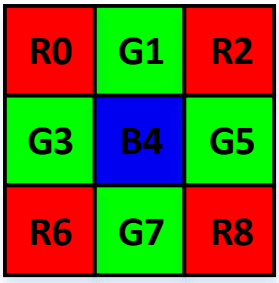
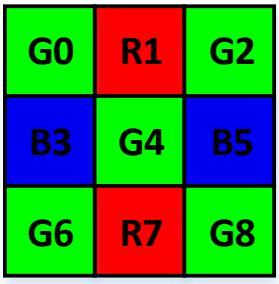
	$R4' = \frac{(R3 + R5)}{2}$ $G4' = G4$ $B4' = \frac{B1 + B7}{2}$
	$R4' = R4$ $G4' = \frac{G1 + G3 + G5 + G7}{4}$ $B4' = \frac{B0 + B2 + B6 + B8}{4}$
	$R4' = \frac{R0 + R2 + R6 + R8}{4}$ $G4' = \frac{G1 + G3 + G5 + G7}{4}$ $B4' = B4$
	$R4' = \frac{R1 + R7}{2}$ $G4' = G3$ $B4' = \frac{B3 + B5}{2}$

Figure 10 – Bilinear demosaicking algorithm (Area scan)

6.1.2.2 Gradient corrected bilinear demosaicking (Line scan)

For line-scan cameras with Bayer filter a special gradient corrected reconstruction is used. The reconstruction forms a single image line out of two lines acquired from camera sensor. The reconstruction uses a gradient corrected interpolation in a 3-by-2 pixel neighborhood.

The interpolation kernel differs for even/odd columns and is according to the Figures shown below. The calculations are performed with full 16 bits resolution.



$$R1' = \frac{R0 + R2}{2} + G1 - \frac{G3 + G5}{2}$$

$$G1' = G1$$

$$B1' = B4$$



$$R1' = R1$$

$$G1' = G4$$

$$B1' = \frac{B3 + B5}{2} + G4 - \frac{G0 + G2}{2}$$

Figure 11 – Bilinear demosaicking algorithm (Line scan)

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6.1.3 Data Packing Mode

Data packing reduces unpacked data padding overhead, thus increasing transfer rates without losing data. Different data packing bitnesses layouts are described as follows:

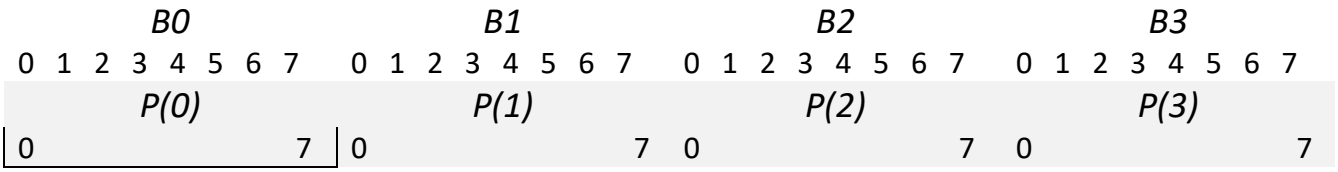


Figure 12 – Packing of 8 bit pixels

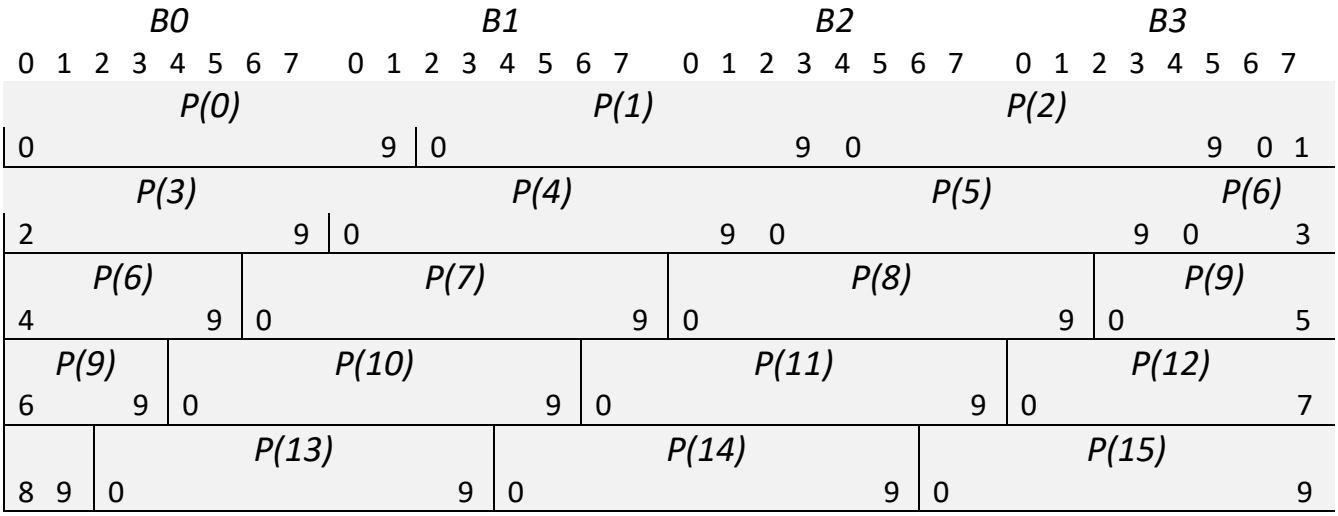


Figure 13 – Packing of 10 bit pixels

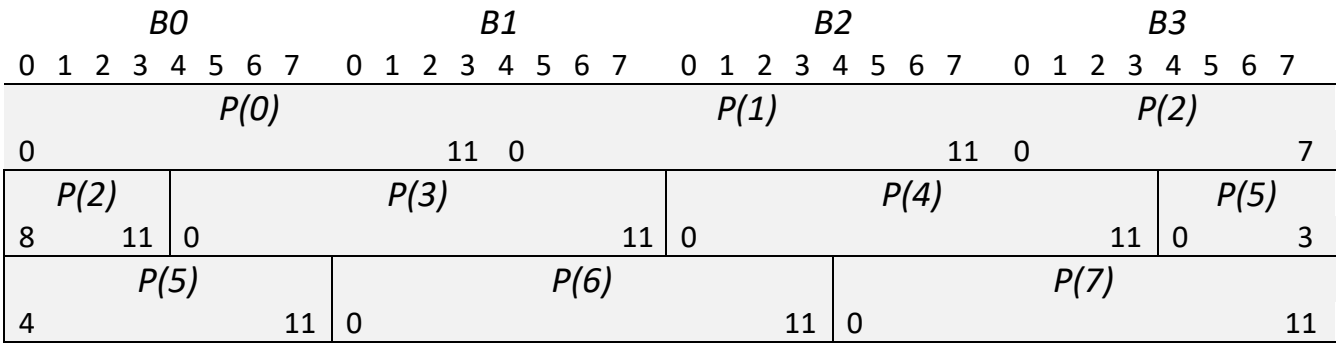


Figure 14 – Packing of 12 bit pixels

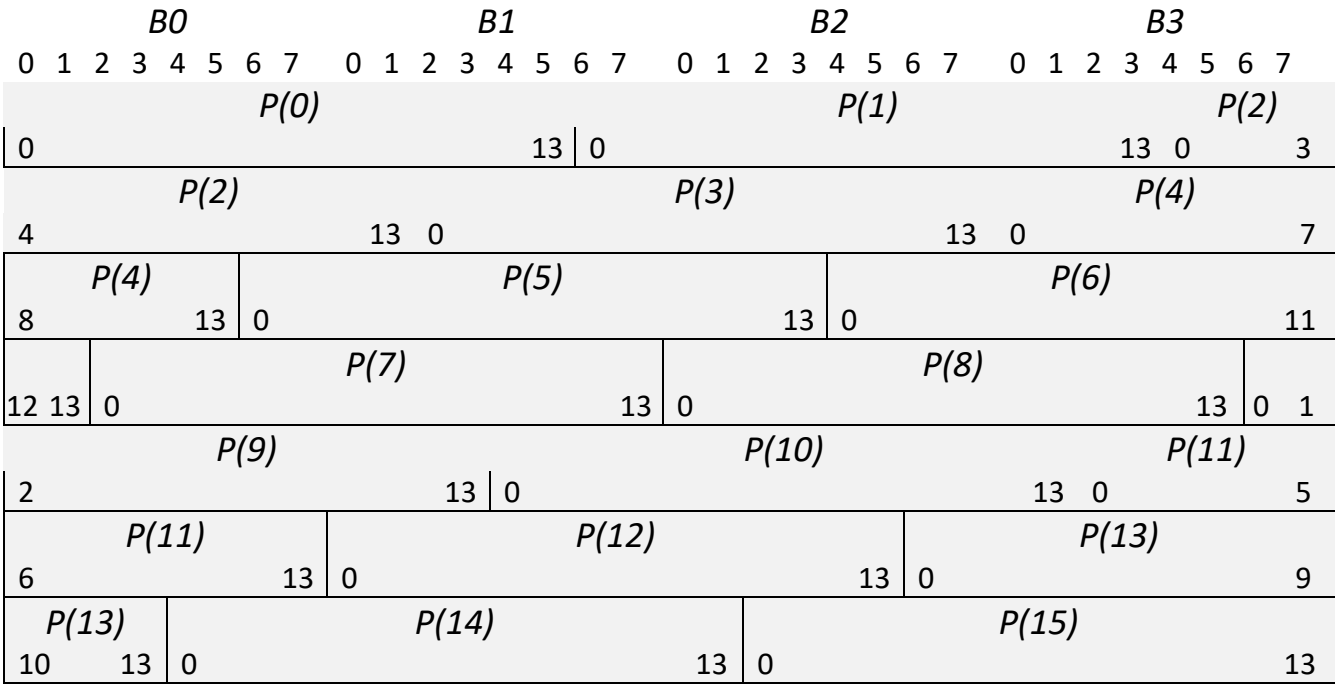


Figure 15 – Packing of 14 bit pixels

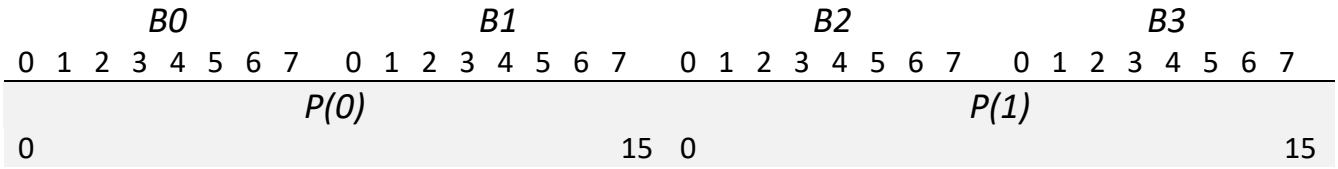


Figure 16 – Packing of 16 bit pixels

By default, stream output will be Unpacked, meaning 10, 12 and 14 bit data will be padded and fit into 2 bytes for each pixel channel. To achieve data packing, “PackedDataMode” should be configured to “Packed_RowAligned32” mode. This will allow to preserve originally packed data or pack an unpacked data stream.

The output stream will be modified as such, that every line will be padded at its end, so byte count will be 32bit aligned. Such approach will accommodate in line manipulation and sequencing.

6.2 Trigger Control

KAYA's Frame Grabbers incorporate a hardware based image processing system that is able to deliver maximum frame rate without effecting system performance. The image processing features includes Bayer demosaic, color transformation matrix, decimation etc. The structure of the image processing pipeline can be seen in the figure below:

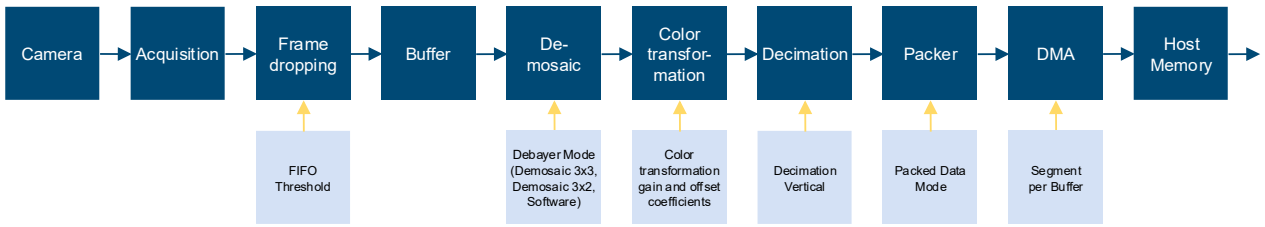


Figure 17 – Hardware based image processing pipeline

The Acquisition triggers are stream oriented; these are issued through internal logic while the system is in data acquisition mode. When configured in this mode, the camera will always stream the images, while Frame Grabber will select which images it should receive based on Acquisition trigger. The flow of the trigger signal in this mode is described in figure below.

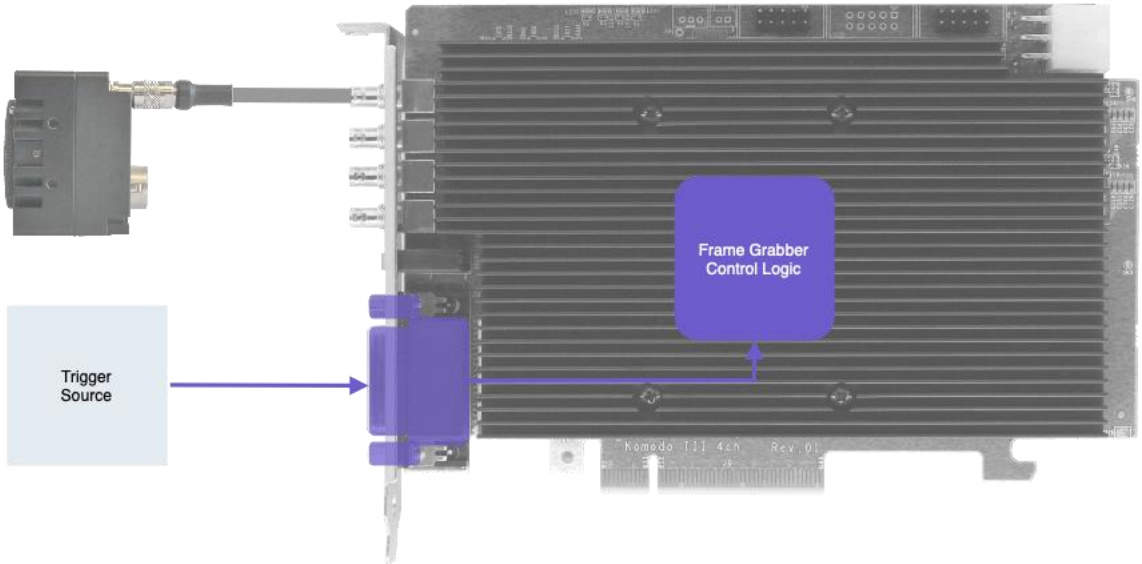


Figure 18 – Acquisition stream trigger source

Internal or external signals/events can act as a source for these triggers. Certain cameras can also be configured to issue triggers for the Frame Grabber over the relevant CoaXPress channel. In some cases, both Camera triggers and Frame Grabber triggers can be used simultaneously to achieve desired effect. Also, a signal can be configured to perform as a trigger for other signals which consequently will be the trigger for Frame Grabber or Camera. The structure of the Acquisition trigger mechanism is described in figure below:

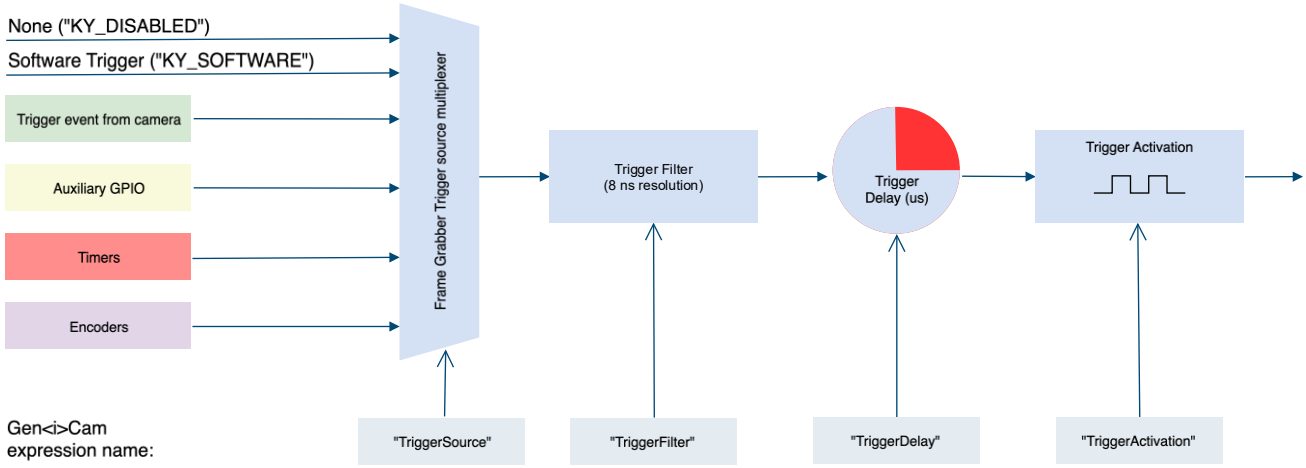


Figure 19 – Acquisition stream trigger structure

The parameters of the acquisition triggers are described in the next table.

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: StreamControl \ TriggerControl						
Trigger Mode	Controls if the trigger is active	TriggerMode [CameraSelector]	Enumeration	0 1	Off On	
Trigger Activation	Specifies the activation mode of the trigger	TriggerActivation [CameraSelector]	Enumeration	0 1 2 3 4	RisingEdge FallingEdge AnyEdge LevelHigh LevelLow	
Trigger Source	Specifies the internal signal or physical input line to use as the trigger source. The selected trigger must have its TriggerMode set to On	TriggerSource [CameraSelector]	Enumeration			See section Link Trigger Source options
Trigger Delay	Specifies the delay to apply after the trigger reception activating it	TriggerDelay [CameraSelector]	Float			In units of microseconds (us)
Trigger Filter	Filter for trigger, helps prevent signal de-bouncing	TriggerFilter [CameraSelector]	Float			In units of microseconds (us) 8ns resolution using fraction value
Trigger Software	Generates an internal trigger	TriggerSoftware [CameraSelector]	Command	1 - Activate		To issue command “TriggerSource” must be set to “Software”
Trigger Event Mode	Select the trigger event generation mode	TriggerEventMode [CameraSelector]	Enumeration	0 1 2 3	Disabled RisingEdge FallingEdge AnyEdge	

Table 27 – Acquisition Triggers parameters

6.2.1 Trigger activation mode

The trigger activation mode configures the capture criteria of signal state. Default value is Rising Edge, which will trigger a frame on signal rising edge event. The different modes functionality is as follows:

6. Any Edge: The frames will be acquired both on rising and falling edges of the trigger source.
7. Rising Edge: The frames will be acquired only on rising edge of the trigger source. Falling edge of the source is ignored.
8. Falling Edge: The frames will be acquired only on falling edge of the trigger source. Rising edge of the source is ignored.
9. Level High: High signal level enables a continuous image acquisition, Low signal level will halt the trigger generation.
10. Level Low: Low signal level enables a continuous image acquisition, High signal level will halt the trigger generation.

6.2.2 Trigger signals filter

The filter of the trigger signals acts as a de-bouncing mechanism for better noise immunity. By default, the filter is disabled with the value of 0. The signal filter resolution can be set at 8ns intervals for high resolution functionality. If the trigger filter is set to a larger value than the width of the trigger pulse, then the pulse will be filtered out and no trigger will occur. Available interface in API provides input in microsecond; nevertheless, to achieve higher resolution, relevant fraction values should be entered after the decimal point.

6.2.3 Trigger Delay

The trigger delay is a mechanism for postponing the incoming signal for a specified number of microseconds. As a result, trigger will be issued after specified time delay to overcome known system latency or set trigger generation period. To disable, value 0 should be set.

6.2.4 Trigger Event Mode

Acquisition trigger event may be enabled for selected camera. This will generate event callback whenever such trigger is generated in hardware. Steps to enable and use such event mechanism are as follows:

1. "TriggerEventMode" is a grabber parameter subordinate to connected camera. Use the KYParametersHandler_SetValue() to set the parameter value to "RisingEdge" for example.
6. Register a callback function for Auxiliary events using KYVPLibExtension library KYVPExtension_PCIInterface_AuxDataCallback_Register() function.
2. To access the data attached to such event please refer to KYVP_AUX_DATA pointer in the callback.

6.2.5 Steps to properly configure Frame Grabber Triggers

7. "TriggerMode" is a Stream parameter subordinate to connected camera. Use the KYVPParametersHandler library function KYParametersHandler_SetValueEnumByValueName() to set the parameter value to "On".
8. The trigger source should be selected according to provided sources and available card GPIO. Only one source can be active, for each camera, at any time.
9. The Trigger Filter resolution ("TriggerFilter"), Activation Mode ("TriggerActivation") and Trigger Delay ("TriggerDelay") parameters should be configured according to desired output.
10. In some cases, the trigger sources should also be configured via provided API before trigger configuration is complete. (e.g if "KY_TIMER_ACTIVE_0" is to be selected as Camera Trigger source, then "Timer0" should first be configured as described in Timer Block configuration in this chapter).
11. After all configurations are complete, start the acquisition. At this point acquisition mechanism will wait for trigger, and Frame Grabber will acquire data upon trigger arrival.

6.3 Statistics and Tests

The stream statistics and tests reflect the state of data flow in the PCI Device for each connected camera. These will be available only after a camera has been discovered and opened. Some parameters represent the quantity and period of received stream packets, while others count errors generated by corrupted data or data overflow. These parameters may be read on each received frame for each camera stream to extract additional information and detect errors on acquisition path. The acquisition stream statistics are summarized in the following table.

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: StreamFeatures \ StatisticsAndTests						
HW Stream Selector	Selects the HW stream to monitoring and control	HWStreamSelector	Integer	>0		
CRC Error Counter	Camera CRC Error Counter. Number of CRC errors generated from corrupted data packets	CRCErrorCounter [CameraSelector]	Integer			Errors are generated from corrupted data packets
RX Packet Counter	Camera RX Packet Counter. Total number of packets received from the camera.	RXPacketCounter [CameraSelector]	Integer			
Drop Packet Counter	Camera Drop Packet Counter. Number of packets dropped due to corruption	DropPacketCounter [CameraSelector]	Integer			
RX Frame Counter	RX Frame Counter. Number of received frames from camera	RXFrameCounter [CameraSelector]	Integer			
Drop Frame Counter	Camera Drop Frame Counter. Number of frames dropped due to corruption	DropFrameCounter [CameraSelector]	Integer			
Drop StreamId Counter	Camera Drop Stream Id Counter. Number of frames dropped due to StreamId corruption	DropStreamIdCounter [CameraSelector]	Integer			
RX Line Counter	RX Line Counter	RXLineCounter	Integer			
RX IH Counter	RX Image Header Counter	RXIHCounter	Integer			
Drop IH Counter	Drop Image Header Counter	DropIHCounter	Integer			
Statistics Counters Reset	Reset all statistics counters	StatisticsCountersReset	Command	1 – Activate		
Frame reception start latency	Latency time in microseconds(us) from frame reception start until user frame acquisition	LatencyFrameStart	Integer			
Frame reception finish latency	Latency time in microseconds(us) from frame reception finish until user frame acquisition	LatencyFrameEnd	Integer			
Latency Counters Reset	Reset all Latency related counters	LatencyCountersReset	Command	1 – Activate		
FIFO Threshold	FIFO threshold, FIFO fill level	FifoThreshold	Integer			
Actual frame rate	Actual acquisition frame rate	AcquisitionFps	Float			

Table 28 –Stream statistics parameters

6.3.1 Frame Acquisition Latency

Latency mechanism provides a criteria to determine time spend processing frame received from camera. Consequently calculates the period passed between the moments the camera has sent a new frame and when user received this data in Host Application. "LatencyFrameStart" holds time value in units of microseconds (usec) computed between the moments when a frame reception start in the Frame Grabber firmware and when user has requested this frame in Host Application. "LatencyFrameEnd" holds time value in units of microseconds (usec) computed between the moments when a complete frame has been received in firmware, and when user has requested this frame in Host Application.

6.3.2 FIFO Threshold

A threshold on a fill level of on-board memory buffers to decide whenever to drop the frames in case the PCIe bandwidth is not enough to transfer the whole image stream. Larger values will result in larger frame latency but in longer frame recording till the dropping starts. A shorter value will result in lower latency but the frame dropping will start sooner. Use this parameter only if the PCIe bandwidth limits your stream, otherwise leave it at default value. Threshold default value is 32MB quantified in Bytes. The threshold value depends on hardware capabilities and mounted memory banks.

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6.4 Transport Layer Control

General settings for data transport (commands and stream) between the PCI Interface (Frame Grabber) and Camera. Parameters may vary depending on the Frame Grabber protocol.

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: TransportLayerControl						
Control Packet Data Size	This provides the control packet data size	ControlPacketDataSize [CameraSelector]	Integer			Units in bytes.
Stream Packet Data Size	This provides the stream packet data size	StreamPacketDataSize [CameraSelector]	Integer			Units in bytes.
Image1StreamID	This gives the Stream ID of the primary image stream from the Device	Image1StreamID [CameraSelector]	Integer			
Image Acquisition Filter Enable	Enables condition filter, under which the image was acquired	ImageAcquisitionFilterEnable	Boolean	0	False	
				1	True	

Table 29 – Transport Layer Control Parameters

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6.5 Image Acquisition filter

Image Acquisition Set byte is part of the CLHS stream data header. This byte may be used to convey camera proprietary information about the condition under which the image was acquired. Image Acquisition Set description can be found in CLHS official document under the “Video Data Message” section.

KAYA’s CLHS compatible Frame Grabber implements the Image Acquisition Filter interface, subject to firmware and software capabilities. When enabled, the mechanism controls the filtration of configured Image Acquisition Set values, thus dropping incompatible frames.

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: TransportLayerControl						
Image Acquisition Filter Selector	Condition filter selector, for specific Image Acquisition Set byte value	ImageAcquisitionFilterSelector	Integer	0 - 255		
Image Acquisition Filter Value	Filter(“Disabled”) or allow(“Active”) frame acquisition with the selected condition value	ImageAcquisitionFilterValue [ImageAcquisitionFilterSelector]	Enumeration	0 1	Disabled Active	

Table 30 – Image Acquisition Filter Control

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6.6 Transport Channel Selector

The Transport Channel Selector indicate the low level connection packets sent between the Host and Device, used for link synchronization. In case of missing or unstable connection, the counters will indicate attempts to resynchronize the link.

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam	
Gen<i>Cam Category: TransportLayerControl						
Revision Packets	Number of Revision packets on selected link of the device	RevisionPackets [CameraSelector]	Integer			
Command Packets	Number of Command packets on selected link of the device	CommandPackets [CameraSelector]	Integer			
FEC Received Packets	Received packets by FEC	FEC_RXPackets [CameraSelector]	Integer			
FEC Corrected Packets	Number of corrected packets by FEC	FEC_CorrectedPackets [CameraSelector]	Integer			
FEC Corrupted Packets	Number of uncorrectable packets	FEC_CorruptedPackets [CameraSelector]	Integer			
Transport Counters Reset	Reset all transport counters	TransportCountersReset	Command	1 – Activate		

Table 31 – CLHS link connection counters

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6.7 Color transformation

The color transformation can be used for color correction operators such as adjusting white balance, color transformation, brightness or contrast. The Color Transformation is a linear operation taking as input a triplet of Components (C0, C1, C2) for a color pixel (Typically: Rin, Gin, Bin representing a RGB color pixel). This triplet is first multiplied by a 3x3 matrix and then added to an offset triplet. The equation is given in the following form:

$$\begin{pmatrix} C0out \\ C1out \\ C2out \end{pmatrix} = \begin{pmatrix} Gain00 & Gain01 & Gain02 \\ Gain10 & Gain11 & Gain12 \\ Gain20 & Gain21 & Gain22 \end{pmatrix} \begin{pmatrix} C0in \\ C1in \\ C2in \end{pmatrix} + \begin{pmatrix} Offset0 \\ Offset1 \\ Offset2 \end{pmatrix}$$

And in particular to RGB images:

$$\begin{pmatrix} Rout \\ Gout \\ Bout \end{pmatrix} = \begin{pmatrix} RR & RG & RB \\ GR & GG & GB \\ BR & BG & BB \end{pmatrix} \begin{pmatrix} Rin \\ Gin \\ Bin \end{pmatrix} + \begin{pmatrix} Ro \\ Go \\ Bo \end{pmatrix}$$

For example, an RGB to YUV conversion of 8bit data can be achieved by the formula below:

$$\begin{pmatrix} Y \\ U \\ V \end{pmatrix} = \begin{pmatrix} 0.299 & 0.587 & 0.114 \\ -0.147 & -0.289 & 0.436 \\ 0.615 & -0.515 & -0.100 \end{pmatrix} \begin{pmatrix} Rin \\ Gin \\ Bin \end{pmatrix} + \begin{pmatrix} 0 \\ 128 \\ 128 \end{pmatrix}$$

6.7.1 Monochrome image special case

A special case of image transformation is applicable for monochrome images to achieve gain/offset operator. For this case the gain matrix should be set to diagonal gain and offset should be the same for each component as below.

$$\begin{pmatrix} C0out \\ C1out \\ C2out \end{pmatrix} = \begin{pmatrix} Gain & 0 & 0 \\ 0 & Gain & 0 \\ 0 & 0 & Gain \end{pmatrix} \begin{pmatrix} C0in \\ C1in \\ C2in \end{pmatrix} + \begin{pmatrix} Offset \\ Offset \\ Offset \end{pmatrix}$$

The color transformation parameters are described in the following table.

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category:	ExtendedStreamFeatures \ ColorTransformationControl					
Color transformation matrix coef RR	Color transformation matrix coef RR	ColorTransformationRR	Float			
Color transformation matrix coef RG	Color transformation matrix coef RG	ColorTransformationRG	Float			
Color transformation matrix coef RB	Color transformation matrix coef RB	ColorTransformationRB	Float			
Color transformation matrix coef R0	Color transformation matrix coef R0	ColorTransformationR0	Float			
Color transformation matrix coef GR	Color transformation matrix coef GR	ColorTransformationGR	Float			
Color transformation matrix coef GG	Color transformation matrix coef GG	ColorTransformationGG	Float			
Color transformation matrix coef GB	Color transformation matrix coef GB	ColorTransformationGB	Float			
Color transformation matrix coef G0	Color transformation matrix coef G0	ColorTransformationG0	Float			
Color transformation matrix coef BR	Color transformation matrix coef BR	ColorTransformationBR	Float			
Color transformation matrix coef BG	Color transformation matrix coef BR	ColorTransformationBG	Float			
Color transformation matrix coef BB	Color transformation matrix coef BB	ColorTransformationBB	Float			
Color transformation matrix coef B0	Color transformation matrix coef B0	ColorTransformationB0	Float			

Table 32 – Color transformation control parameters

6.8 Meta Data Control

When the Metadata insertion feature is activated, some Metadata information will be delivered along or instead to the data stream. The inserted information will be configured according to selected Metadata insertion mode.

6.8.1 Mode1

When the Metadata insertion Mode1 is activated, the first 12 bytes of each image line are replaced by a fixed set of metadata information as follows:

- The logical state of System I/O input lines
- The value of the motion encoder pulse counter
- The value of the Camera Link LVAL pulse counter

Bit	Function
0	OptoCoupled Input 0
1	OptoCoupled Input 1
2	OptoCoupled Input 2
3	OptoCoupled Input 3
4	OptoCoupled Input 4
5	OptoCoupled Input 5
6	OptoCoupled Input 6
7	OptoCoupled Input 7
8	LVDS Input 0
9	LVDS Input 1
10	LVDS Input 2
11	LVDS Input 3
12	TTL 0
13	TTL 1
14	TTL 2
15	TTL 3

Table 33 – System I/O input lines

Format of the metadata for each line in bytes:

3:0	7:4	9:8	11:10	EOL:12
Camera Link LVAL pulse counter (32 bit Little Endian) The counter resets on start of acquisition	Motion encoder 0 pulse counter (32 bit Little Endian)	Logical state of System I/O input lines (16 bit Little Endian)	Reserved	Video raw data

Table 34 – Metadata format

* EOL – End Of Line

The metadata control parameters are described in the table below.

Parameter	Description	Gen<i>Cam name	Type	Possible values		Remarks
				Value	Gen<i>Cam name	
Gen<i>Cam Category: ExtendedStreamFeatures \ MetaDataControl						
Meta Data Enable Mode	Inserts metadata information according to selected mode	MetaDataMode	Enumeration	0 1	Disable Mode1	

Table 35 – Metadata control parameters

7 Multiple Frame Grabber Synchronization

In order to synchronize multiple Frame Grabbers together the following should be done:

1. The Frame Grabbers must be connected together with a harness.
2. One of the Frame Grabber will be defined as master and configured to provide timer pulses to other slave Frame Grabbers.
3. Connected cameras should be set to Triggered mode (camera vendor dependent).
4. Camera Trigger parameters should be enabled in all Frame Grabbers to provide triggers to cameras.

In order to achieve a synchronized triggering to all the cameras a sync harness is connected to J1 of all the Frame Grabbers. The wiring diagram of the harness can be seen in the diagram below.

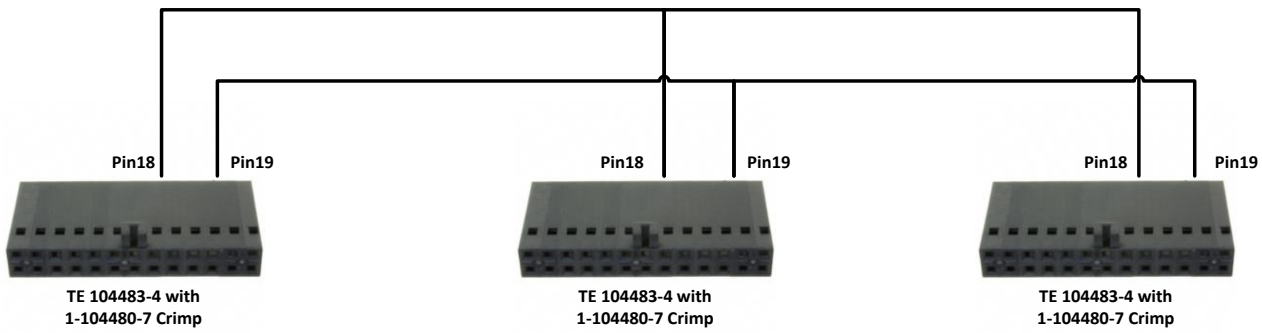


Figure 20 – Frame Grabber synchronization wiring

One of the Frame Grabbers operates as master and others as slaves for camera triggering. The example above shows configuration of three Frame Grabbers for 90Hz frame rate. If other frame rates are needed, the “TimerDelay” and “TimerDuration" values should be adjusted accordingly.

The configuration sequence includes the following steps:

1. Configure timer to generate 90Hz waveform on master card. Please refer to Timer Control section for timer description.
2. Configure trigger path for each camera by calling KYParametersHandler_SetValueEnumByValueName() (for parameters of Enumeration type) and KYParametersHandler_SetValue() (for parameters of other type) with camera handle. Please see 5.2 for detailed description.
3. Configure GPIO to synchronize between different boards. Please refer to **Error! Reference source not found.** s ection for detailed description.

The Frame Grabbers in the control PC should be configured in the following sequence:

Gen<i>Cam Name	Type	Card 0 Value (Master)	Card 1 Value	Card 2 Value	Comment
TimerSelector	Enumeration	“Timer0”	NA	NA	
TimerDelay	Float	5555.55	NA	NA	Half cycle for 90Hz
TimerDuration	Float	5555.55	NA	NA	Half cycle for 90Hz
TimerTriggerSource	Enumeration	“KY_CONTINUOUS”	NA	NA	
CameraTriggerMode	Enumeration	“On”	“On”	“On”	For each camera
CameraTriggerActivation	Enumeration	“AnyEdge”	“AnyEdge”	“AnyEdge”	For each camera
CameraTriggerSource	Enumeration	“KY_TTL_0”	“KY_TTL_0”	“KY_TTL_0”	For each camera
LineSelector	Enumeration	“KY_TTL_0”	“KY_TTL_0”	“KY_TTL_0”	
LineMode	Enumeration	“Output”	“Input”	“Input”	
LineSource	Enumeration	“KY_TIMER_ACTIVE_0”	“KY_DISABLED”	“KY_DISABLED”	

Table 36 – Frame Grabber required settings

REFERENCES

Supported vision standards:

CoaXPress GEN<i>CAM

Vision Point documentation:

Vision Point

TECHNICAL SUPPORT AND PROFESSIONAL SERVICE

If you searched the documents and could not find the answers you need, contact KAYA Instruments support service:

- Create a support request on the web: support.kayainstruments.com
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Visit us at kayainstruments.com for comprehensive information.

SUBMITTING A SUPPORT REQUEST

When opening a support request, please provide the following information when applicable:

For Frame Grabbers:	For Camera:	For Range Extender:
• Vision Point Diagnostic Info* • Serial number of Frame Grabber • Camera model • SFP+ module model • CoaXPress/Fiber cable model and length • External power or PoCXP • PC motherboard model *In the Vision Point app, use menu option Help > Collect diagnostic info.	• Vision Point Diagnostic Info (or frame grabber being utilized) • Serial Number of Camera • XML dump and/or description of how the camera is being utilized • Description of issue • SFP+ module model • CoaXPress/Fiber cable model and length • External power or PoCXP	• Range Extender Model • Serial Number of Range Extender • SFP+ module model • CoaXPress/Fiber Cable model and length • PC configuration • Operating System • Software Revision • Camera and Frame Grabber Manufacturer and Model

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